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BETWEEN SUSTAINABILITY CLAIMS AND ACTUAL PERFORMANCE

Measuring Greenwashing Risk in CEO Letters and Its Relation to ESG Ratings and Stock Performance Among Nordic B2B Firms

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ABSTRACT

As sustainability expectations on listed firms continue to grow, so does the risk that corporate environmental communication becomes more rhetorical than substantive. This study investigates whether greenwashing risk embedded in CEO letters is associated with lower ESG ratings and weaker stock returns among Nordic listed business-to-business firms over the fiscal years 2022-2024. The Nordic context is chosen particularly given the region's reputation for sustainability leadership, which may create stronger incentives for firms to align their corporate narratives with sustainability expectations.

To address this, the study develops the Greenwashing Warning Risk Index (GWRI), a novel sentence-level measurement instrument grounded in signaling theory, decoupling theory, and impression management theory. The index is applied to a panel of 113 Nordic B2B firms yielding 339 firm-year observations. Greenwashing risk scores are regressed against Refinitiv ESG ratings and forward-looking stock returns with industry and year fixed effects, validated across multiple model specifications. The GWRI captures five sentence level dimensions including vague claims, forward looking deflection, and selective disclosure. The tool was constructed through a theory-driven coding approach and validated by reliability testing.

The findings are asymmetric. Higher GWRI scores are significantly and negatively associated with ESG ratings, with a one standard deviation increase in greenwashing risk corresponding to approximately 4.9 rating points lower ESG score. No statistically significant association is found between greenwashing risk and stock returns. These results suggest that ESG rating agencies, as analytical intermediaries with structured text-sensitive methodologies, detect linguistic sustainability signals in CEO letters, while investor remain largely insensitive to this dimension of corporate narrative. The divergence may reflect the opaque nature of CEO letter rhetoric for retail and institutional investors who lack the analytical tools to systematically evaluate textual sustainability signals. The study concludes that the consequences of greenwashing rhetoric are concentrated among the intermediaries equipped to identify it, with implications for sustainable investment practice, the design of ESG disclosure regulation, and the development of more linguistically sensitive frameworks in future greenwashing research.

Keywords: Greenwashing, ESG Ratings, CEO letters, Nordic B2B firms, Corporate Sustainability Disclosure, Greenwashing Warning Risk Index (GWRI), Stock Returns

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List of Abbreviations

B2B	Business-to-Business
B2C	Business-to-Consumer
CSR	Corporate Social Responsibility
CSRD	Corporate Sustainability Reporting Directive
EMH	Efficient Market Hypothesis
ESG	Environmental, Social, and Governance
GWRI	Greenwashing Warning Risk Index
LLM	Large Language Model
LSEG	London Stock Exchange Group
MSCI	Morgan Stanley Capital International
NGO	Non-Governmental Organization
NLP	Natural Language Processing
PCA	Principal Component Analysis

SFL	Systemic Functional Linguistics
VADER	Valence Aware Dictionary and Sentiment Reasoner

1. Introduction

This chapter begins with an explanation of the background of the topic. Further, it presents the problem formulation, which forms the basis for the purpose of the study, as well as the research questions and limitations.

1.1 Background

Over the past decades, the global economy has faced a period of high uncertainty and structural changes caused by global events such as climate change, geopolitical instability, and rapid technological transformation. These events have demonstrated weaknesses in current economic models and highlighted the need for a more sustainable corporate framework (IPCC, 2023, pp. 5-6). This change in focus has made it easier to incorporate sustainability into corporate strategy, and now sustainability is no longer a marginal concern, but also a strategic priority for firms seeking long-term business planning and stakeholder engagement. Simultaneously, all listed companies are increasingly expected to combine economic performance with environmental stewardship, that aligns with both regulatory requirements and the expectations of diverse stakeholders (Shmelev & Gilardi, 2025, p. 1).

In order to understand how this shift is communicated, one must look at the evolution of the annual report. In the past few decades, corporate reporting has expanded beyond equity holders to include employees, regulators, and the public, evolving from a statement of accounts into a sophisticated communication tool that combines financial data with strategic corporate storytelling (Merkl-Davies & Brennan, 2007, pp. 116-119). Its primary purpose was stewardship, providing a "true view" of a firm's accounts to owners and investors not involved in daily operations, to ensure that invested capital was not being misused by management. As corporate audiences broadened, so did the communicative ambition of these documents, and the narrative sections of annual reports became an increasingly important vehicle through which management constructs and projects a public image of the firm (Merkl-Davies & Brennan, 2007, pp. 116-119).

Today, the annual report is the primary tool for both financial and non-financial data, including Environmental, Social, and Governance (ESG) metrics. While its legal purpose remains focused at the disclosure of financial health, its strategic purpose has shifted towards building legitimacy and trust (Eccles & Krzus, 2010, pp. 231-233). At the same time, with the growing amount of information, it becomes particularly difficult for investors and other stakeholders to reliably assess sustainability performance (WFE, 2019, pp. 3-4). The disclosure of environmental and social information in corporate reports often reflects stakeholder demands, but the lack of uniform standards and assurance mechanisms creates uncertainty regarding the accuracy and comparability of reported information (ISO, 2024). Within the dense, often hundreds of pages of a modern annual report, the CEO's letter (or "Letter to Shareholders") stands out as a unique and critical artifact and it plays a particularly important role in addressing stakeholders directly (Arvidsson, 2023, pp. 27-53). Previous research consistently identifies that CEO letters are one of the most widely read and influential sections of the entire document (Na et al., 2020, p. 220; Mäkelä & Laine, 2011, pp. 217-231). Unlike the rigid, audited tables of the financial statements, which are bound by strict accounting rules, the CEO letter offers a narrative space. This is where executives provide a personal perspective on the firm's trajectory and frame future strategic goals in a language that is accessible to a broad audience (Yuthas et al., 2002, pp. 141-144). It is this specific lack of formal auditing that grants management significant discretion in how they craft their message (Boudt & Thewissen, 2019, pp. 77-115;

Davis et al., 2012, pp. 846-868). These letters are often forward-looking, highlighting long-term initiatives and plans rather than simply describing past results, making them a key source of information for investors and analysts assessing the company's outlook (Yan et al., 2019, pp. 465-492).

However, Merkl-Davies and Brennan (2013, pp. 110-112) argue that the high degree of managerial discretion in corporate narratives introduces the risk of impression management, which may not reflect underlying economic performance. At its core, impression management is a psychological and sociological process where individuals or organizations attempt to control the perceptions others form of them (Goffman, 1959). As the importance of sustainability increased, it has made the ESG communication a mandatory component in CEO letters, where environmental and social initiatives are frequently described by executives, positioning their companies as responsible corporations (Arvidsson, 2023, pp. 27-53). This pressure can lead to "symbolic management", where the firm uses language that align itself with social values without necessarily changing its underlying operations behavior (Hess, 2008, p. 465). When this gap between narrative and reality becomes significant, it manifests as greenwashing, which from a theoretical perspective, commonly characterized as the practice of making exaggerated or inaccurate claims about environmental or social performance supporting them by substantive actions (Flammer, 2021, p. 500; Shimamura et al., 2025, p.1). Greenwashing can occur in many forms, for example a company's public image may be separated from its actual environmental impact, or they can highlight one small eco-friendly project to distract from a much larger, more damaging environmental footprint. These practices often evolve the use of vague language, omission of negative outcomes, optimistic language or future oriented statement without any concrete timelines or measurable benchmarks. These practices do not only confuse investors and other stakeholders but also erodes the trust that the annual reports are supposed to build (Lyon & Montgomery, 2015, p. 224; Walker & Wan, 2012, p. 227), making it harder for truly sustainable firms to distinguish themselves from those who just market themselves as "green".

Recent studies have demonstrated that specific linguistic choices such as complexity of the text, the positivity of the tone, and narrative framing in sustainability disclosures, can serve as "red flags" and is strongly correlated with greenwashing tendencies (Hu et al., 2024, pp. 1-19). Furthermore, because Abu Bakar & Ameer (2011, pp. 50-58) have shown that there is a positive correlation between the readability of the corporate narratives and profitability, companies feel a strong pressure to "polish" the wording of their reports. This underscores the critical need for rigorous textual analysis to determine if an executive's sustainability message is a reliable signal of future value or a misleading distraction that could lead to poor subsequent market performance.

1.2 Research Gap

ESG disclosures and greenwashing have been researched significantly due to the rising importance of the topics around the world. However, most of the existing research focuses on empirical evidence from CSR/sustainability reports or ESG disclosures. According to recent research, the assurance for environmental disclosures lacks specific mandatory requirements which can raise concerns about truthfulness of the information. Additionally, the voluntarily disclosures pose risk of misleading the readers for strategic purposes and related stakeholders lacks the ability to determine the sincereness of sustainability commitments. For this reason, greenwashing is originally a phenomenon based on disclosures which can occur if the

companies select disclosing positive sustainability related information meanwhile not disclosing negative performances equally (Bernini & La Rosa, 2024, pp. 413-417). Hence, the readers get an overly positive impression of sustainability-related outcomes.

We identified that executive messages or CEO letters have received significantly less attention in the context of greenwashing even though it is a highly valuable and influential communication channel that companies use. As mentioned previously, executive messages are the most influential and read section of the annual report. It is also considered to be representation of sign of priorities and strategies of the company as the narrative focuses on operations, corporate vision and values. Mentioning sustainability related topics in the message signal that sustainability issues are high up on their corporate agenda (Arvidsson, 2023, pp. 27-34). On the other hand, readers can struggle identifying the sincerity of the CEO message because the quality of the disclosure can vary, and the executives can use the message in a misleading manner. This situation creates a meaningful gap both academically and managerially.

Additionally, we analyzed that the identified research gap becomes more critical in B2B settings. B2B markets are special as sustainability claims are not only for public reputation but also important for long-term relationships with business partners. According to Walker & Wan (2012, p. 231), business relationships are especially vulnerable to greenwashing because the partners can withdraw if a firm is perceived to be untrustworthy or opportunistic. Hence, the consequences are high if the firm wins businesses in the short run with misleading sustainability claims. However, it remains unclear if sustainability claims in executive messages in B2B firms align with external ESG performance assessment and whether there are any consequences in the stock market performance.

The Nordic region is particularly relevant for studying greenwashing risk in executive communications since it is consistently identified as the global frontrunner in sustainability and CSR. The region has strong institutional structures, stakeholder engagements, and a regulatory environment that progresses continuously. (Gjølberg, 2010, pp. 203-209; Strand et al., 2015, p.1). Hence, sustainability commitments are subject to strict scrutiny. If the market detects deviations on actual performance from rhetoric performance, then the reactions can be more visible and consequential compared to other markets. Moreover, ESG reporting is advanced in the Nordic region, and it increases the availability of third-party ESG ratings. This is an empirical prerequisite for a building a valid greenwashing measure (Khatri & Kjærland, 2023, pp. 2-3).

1.3 Research Purpose

As sustainability and ESG issues become increasingly important to investors, regulators, and other stakeholders, firms face pressure to communicate their environmental and social performance in their narrative disclosures. However, the variety of reporting standards creates opportunities for executives to exaggerate sustainability performance in their letter, which is widely known as greenwashing. Therefore, the aim of this study is to investigate the extent to which greenwashing risk is present in CEO letters of publicly listed B2B firms in Nordic countries, and how it aligns with the third-party agency ESG ratings and financial market performance.

1.3.1 Research Questions

This research focuses on two key questions:

1. *How does the greenwashing risk in executive messages relate to third-party ESG ratings of B2B Nordic firms?*

2. *How do these executive sustainability messages, and any associated greenwashing risk, affect subsequent stock market performance?*

1.4 Delimitations

The scope of the following research is limited to publicly listed firms in the Nordic countries, including Sweden, Norway, Denmark, and Finland. The Nordic region is suitable for this research because stakeholder expectations regarding sustainability are particularly strong and tightly linked to business practices (Strand et al., 2015, p. 1). The chosen industry focus ´business-to-business (B2B) firms, defined as companies that primarily sell their products or services to other businesses rather than individual consumers (Pfoertsch & Scheel, 2012, p. 265). It should be noted that the sample is not restricted to purely B2B firms; rather, it includes firms whose business model is predominantly B2B. Some companies may have a B2C component in their operations, but as long as their core activities are business-to-business, they are retained in the sample. Firms that are primarily B2C are excluded to maintain consistency across observations. In B2B settings, sustainability is often a key factor in business relationships: corporate and public buyers frequently require suppliers to meet certain sustainability standards, such as emissions reporting or certifications. This gives firms strong incentives to present themselves as responsible actors in CEO communications in order to secure long-term contracts (Apostol et al. 2021.

In terms of data, the study focuses on the sustainability-related content of CEO letters, as this is the narrative component most directly comparable to third-party ESG ratings. The time period covered is fiscal years 2022-2024, with stock returns extending through 2025 to capture subsequent market responses to CEO letters published in the previous year. This three-year window captures the most recent developments in sustainability reporting while ensuring availability of all three data types: CEO letters, ESG ratings, and stock prices, for the firms in our sample.

1.5 Outline of the thesis

After the introductory chapter, chapter 2 will present and discuss literature review including previous research on the chosen topic, theories and frameworks as well as the hypothesis of the study. Following, chapter 3 will motivate the methodology approach, discuss the validity and potential limitations and risks. Further, chapter 4 will present the results of the findings, following chapter 5 discussing and analyzing these findings. Finally, chapter 6 will conclude the study highlighting the contributions, ethical and societal implications, limitations, and recommendations for future research.

2. Theoretical Framework and Literature Review

This chapter provides the reader with a theoretical framework including agency theory, signaling theory, and impression management theory. Furthermore, it will introduce the systemic functional linguistics and appraisal framework. It also includes the concept of greenwashing and CEO letters in annual reports, and previous research on examining the ESG performance and risk of greenwashing. These are aimed at building the foundation for our analysis.

2.1 Sustainability Talk in CEO Letters

CEO letters are narrative communication often placed at the beginning of the annual report, to address shareholders and other stakeholders, and it has long been viewed as the most read and influential parts of the report (Na et al., 2020, p. 220; Mäkelä & Laine, 2011, pp. 217-231). Compared to the required and audited sections of the annual report, the idea of including CEO letters and any voluntary disclosures is relatively new, but its use and popularity have rapidly increased in the past decade (Davis et al., 2012, pp. 846-868). One factor contributing to this development, according to Ahmed & Courtis (1999, p. 55), is stakeholders' and investors' discontent with required financial disclosures, which has led them to want additional information on strategy and performance.

It is stated that CEO letters can have the ability to impact corporate operations and practice, as well as business society. According to Amernic & Craig (2004, p. 327), CEO's rhetorical influence is particularly important when organizational legitimacy is under jeopardy, and they are expected to exercise rhetorical leadership. The previous years are thought to have demanded some rhetorical power from leaders in their sustainability talks. Thus, CEO letters, which were formerly centered on market position, financial performance, and shareholder value, are now recognized as crucial tools demonstrating how sustainability is incorporated into the core of the company (Yan et al., 2019, pp. 465-492). This change is in line with stakeholders' increasing demands for responsibility and transparency as well as focus on long-term value development. Despite the paucity of prior research examining the actual CEO talk on sustainability, Albertini et al. (2021, p. 687) discovered that CEOs highlight environmental capital as a significant contributor in their letters. On the other hand, Barkemeyer et al. (2014, p. 254) found that CEO remarks regarding sustainability are just used as an excuse because they are not held accountable for performance linked to sustainability. de-Miguel-Molina et al. (2019, p. 52), also point out that the social aspects of sustainability in CEO letters had an impact on corporate investments decisions.

As previously mentioned, there are fewer restrictions on the content of CEO letters. According to Arvidsson (2019, p. 164), corporate sustainability narratives presented in CEO letters have come under fire for being little more than greenwashing. Greenwashing, according to Ghitti et al. (2024, p. 907), is an agency-driven practice in which managers purposefully reveal environmental information to enhance the company's reputation or satisfy stakeholders without making any significant changes. For example, a firm may highlight minor sustainability projects, such as recycling programs or carbon offset initiatives in CEO letters or sustainability reports while continuing to use environmentally damaging production methods. This selective disclosure may give stakeholders a false impression that environmental performance is strong. According to a recent study by Arvidsson (2023, pp. 50-53), corporate sustainability goals are increasingly communicated through CEO letters. More current communications highlight sustainability as a component of long-term value creation and company strategy, whereas

earlier CEO letters primarily framed sustainability in terms of environmental compliance. However, the researcher warns that the increasing prevalence of sustainability communications in those letters may not accurately reflect the integration of sustainability within companies, emphasizing the importance of critically assessing such disclosures (Arvidsson, 2023, pp. 50-53).

2.2 Information Asymmetry and Agency Theory

Agency theory is one of the fundamental concepts in corporate governance, rooted in the literature of economics. The concept was first introduced by Jensen & Meckling (1976) and focuses on the relationship between a company's managers and shareholders. This creates a scenario where the manager effectively functions as an agent while the shareholder acts as a principal. The heart of the theory addresses issues that arise when the principal assigns a task or piece of work to an agent. However, the core idea of the theory is that agents may not exhibit behavioral alignment with principals due to conflicting goals, varying risk tolerance, or insufficient information. This misalignment creates "agency problems", where managers often prioritize short-term profit or a polished public image over long-term investments such as environmental sustainability (Jensen & Meckling, 1976, pp. 308-309; Eisenhardt, 1989, pp. 58-60; Gryglewicz et al., 2020).

Since shareholders are not involved in day-to-day operations by the company, they constantly strive to make sure that management is genuinely operating in the most beneficial way. This is further supported by Hartwig (2018, p. 186), who argues that there is a risk that management will act opportunistically in its own interest than the interest of the shareholders. For example, the CEO can emphasize short-term profits or achievement while downplaying long-term risks. This occurs because the CEO has more information than shareholders creating an information asymmetry that lies at the heart of agency problems (Omar et al., 2017, p. 2).

Information asymmetry can be described as a lack of knowledge or information among shareholders, but it can also mean that some have a better grasp of the state of the market and have distinct skills and backgrounds to process the information (Akerlof, 1970; Scott, 2015). To reduce information asymmetry, shareholders often employ monitoring mechanisms including audits, financial reporting and, other oversight activities (Hartwig, 2018, p. 186). These measures are referred to as agency costs, and it can be assumed that it is difficult from the principal's perspective to determine how many measures they should use to reduce these agency costs. The assessment of how management's opportunistic behavior may affect principal can also be seen as a potential agency cost (Cotter et al., 2011; Deegan & Unerman, 2011, p. 263).

Expanding upon the agency's perspective, recent research has also connected agency problems to the phenomena of greenwashing. Managers can affect how sustainability performance is conveyed since they have greater in-depth knowledge of the company's internal operations than shareholders or other external stakeholders. This information disparity reinforces the information asymmetry between agents and principals by enabling managers to choose to highlight environmental beneficial efforts (Ghitti et al., 2024, p. 927). Furthermore, the study highlights that governance arrangements meant to reduce agency conflicts may not necessarily eradicate greenwashing. In some circumstances, businesses may still utilize symbolic sustainability communication if they have stronger governance practices or are under greater pressure to demonstrate environmental stewardship. This suggests that even during monitoring

measures, managers may have significant power over how sustainability performance is conveyed to external stakeholders. Thus, greenwashing may persist as an indication of agency problems in corporate sustainability reporting. Importantly, the authors find that businesses that engage in more greenwashing tend to have lower firm values over time, indicating that misleading environmental disclosures may eventually erode investor trust and negatively affect long-term shareholder wealth (Ghitti et al., 2024). This is further supported by research examining the correlation between agency conflicts and stock performance. In a study, Chen et al. (2023, pp. 9-14), finds that agency-related conflicts driven by managerial incentives are associated with less subsequent stock return. According to their results, corporate performance and market valuation may suffer when managers adopt methods that grow the company for their own benefits rather than the benefit of shareholders. This is due to investors' propensity to undervalue companies whose management practices are not entirely in line with the interest of shareholders. Consequently, agency issues negatively affect firm valuation and long-term performance (Chen et al, 2023, pp. 9-14). In the empirical analysis of this study, agency theory provides the conceptual backdrop for interpreting why greenwashing rhetoric persists in CEO letters even when monitoring mechanism exist.

2.3 Signaling Theory in CEO communication

Signaling theory is one of the fundamental approaches in economics and information theory, which explains how, in conditions of information asymmetry, one party conveys a meaningful message to another. This becomes relevant in this research if companies are not fully transparent with the sustainability information they disclosure. Signaling theory was originally introduced by economist Micheal Spence (1973) as part of his study of labor markets and information asymmetries between workers and employers. The theory has been famously adopted in other fields, including corporate finance, marketing, and communications (Connelly et al., 2011, p. 39). In an economic setting, signaling theory discusses how market participants can reduce information asymmetry when two parties have access to different information. Furthermore, in a corporate context, managers, as the senders, who have more complete information, can choose how and what to communicate, and the recipient interprets the signal. (Connelly et al., 2011, p. 39, Berrone, 2017, p. 365; Dumay, 2016, p. 171; Michelon & Parbonetti, 2012, p. 504). Connelly et al. (2011, pp. 39-41) further discusses the cost of signaling, which implies that certain kinds of messages come with a cost. For instance, when a company disclose their forecast for the next year, they incur costs in the form of reputational threats. To put it another way, signaling theory implies that the companies want to announce their goals and achievements in order to show that they are trustworthy.

Among all corporate communication documents, the CEO letters are perhaps the most personal and flexible signaling tool available, as it stands out for the freedom of expression and strategic communication. Because these narratives are so visible and difficult to standardize, it gives room for CEOs to craft unique signals about their sustainability and governance (Brennan & Merkl-Davies, 2013, pp. 109-114). Therefore, it comes with a great weakness in terms of signaling integrity, where managers can emphasize positive results while downplaying environmental impact. This is where the connection between signaling theory and greenwashing arises, which highlights the risk that voluntary narratives such as CEO letters may misrepresent actual environmental performance. Consequently, this gap between rhetoric and actual performance can be best interpreted through decoupling theory. Originally introduced by Meyer & Rowan (1977, pp. 356-359), decoupling theory states that companies do adopt formal policies, commitments, and narratives in order to gain legitimacy while their

business model remains unchanged. As a result, symbolic commitments may become disconnected from substantive environmental performance, increasing the risk that CEO letters function primarily as legitimacy-seeking communication rather than as accurate reflections of corporate behaviors. Thus, the credibility of these narratives is central to their value: when sustainability claims are not supported by concrete timelines, measurable evidence, or verifiable facts, the signal loses its informational content, and the firm may face a market discount once stakeholders recognize the disconnect (Connelly et al., 2011, p. 39; Bernini & La Rosa, 2024, p. 425).

In order to prevent greenwashing in CEO letters, the signaling environment relies on the use of signal verifiers. These are the third-party entities, such as independent auditors or in this case, the ESG rating agencies, that can provide an objective assessment of the company's statements. Zhang et al. (2025, p. 3) argue that within signaling theory, such verification increases the cost of misleading and makes it harder for low-quality companies to copy high-quality ones. ESG ratings restrict managerial discretion and increase the reliability of sustainability signals in capital markets since they employ external data sources and specify standards that go beyond voluntary narratives (Zhang et al., 2025, p. 9). For example, if a CEO's letter is full of heavy and ambitious environmental claims that are later contradicted with poor ESG ratings, the disparity serves as a "counter-signal" alerting the market to possible greenwashing. As noted by Lewin & Warren (2025, p. 421), companies are becoming increasingly conscious of the fact that their narrative signals are being cross-referenced with external information. As a result, a lot of companies today use websites and social media to continuously and symbolically reaffirm their validity. However, potential of signal overload in which stakeholders find it challenging to distinguish between signals that are genuine and those that are merely a component of an impression management, rises with the volume of communication (Lodhia et al, 2020, p. 614).

From a capital market perspective, the signals embedded in corporate disclosure directly shape investor expectations and, by extension, drive stock price movements. Signals perceived as authentic can strengthen investor confidence and bolster a firm's valuation, while those flagged as greenwashing risk triggering sharp market corrections once the underlying inconsistencies come to light (Affolter et al., 2024, p. 7; Itan et al., 2025). Signaling theory therefore indicates that companies can lower their risk of share price collapse by ensuring that their sustainability narratives are credible and substantiated, as investor preferences increasingly incorporate sustainability considerations into valuation (Harmadji et al., 2018, p. 648). Signaling theory provides a theoretical basis for distinguishing between substantive sustainability commitments and vague, aspirational rhetoric in the analysis of CEO letters.

2.3.1 Impression Management in Corporate Disclosures

Closely related to signaling theory, impression management focuses on how organizations strategically shape corporate communications in order to influence how stakeholders perceive their performance and legitimacy. Impression management theory has emerged as an important framework for understanding corporate behavior since the 1960s. Originally, the concept was rooted in social psychology, that was used to describe how individuals curate their behaviors to illustrate a specific public image and please the target audience. Erving Goffman (1959) introduced this concept within the sociological framework of everyday social interactions, describing how individuals perform roles to control how they are perceived by others. To maintain social acceptance, people often tend to try to control the information about themselves to present for the audience (Raban et al., 2017). When this concept is scaled up, it becomes

“Organizational Impression Management”. In this setting, impression management refers to all acts and strategies designed and implemented to influence stakeholders’ perceptions about the company’s health and ethics, and it can be something they do intentionally and unconsciously. Brennan & Merkl-Davies (2013, pp. 110-112) define organizational impression management as the strategic shaping of disclosures to influence stakeholders’ perceptions. This implies that a gap may arise between companies’ underlying actions and how these actions are presented externally. Companies shape information and their image to influence stakeholders' perceptions and views of the company by controlling what information is disclosed and how.

In the sphere of corporate reporting, impression management can be frequently found in press releases, sustainability reports, and particularly in the CEO letters. In the context of sustainability, as demand for ESG information has become significantly important for investors, authorities and business partners, companies now have a strong incentive to use impression management (Cho et al., 2015, p. 80). This increasing pressure from stakeholders has led to the phenomenon of greenwashing becoming more prevalent within organizations today. Nowadays, greenwashing is conceptually interconnected with this concept since many businesses implement these strategies to influence stakeholder perceptions of their environmental performance. In this context, impression management is often done through language framing, selective disclosure, and narrative emphasis (Delmas & Burbano, 2011, p. 65).

Previous empirical studies further support the close relationship between impression management and greenwashing in ESG related disclosures. For example, Sneidereine & Legenzova (2025, pp.1-22) demonstrate that companies use this approach more aggressively in sustainability reports than in traditional management reports. Greenwashing is represented by systematic variations in tone and narrative strength, and according to their findings, greenwashing is not a transient reaction to regulatory pressure, but rather a conscious application of perception management strategies. Similarly, Cho et al. (2015, pp.80-81) shows that sustainability reporting often functions as a symbolic façade, with firms emphasizing positive ESG initiatives while minimizing or omitting negative outcomes. Their analysis suggests that companies strategically manage impressions to maintain legitimacy with stakeholders, even when actual environmental or social performance is limited. Furthermore, Walker & Wan (2012, pp. 227-240) indicated in the study that symbolic actions (“green talk”) are negatively related to financial performance, while substantive actions (“green walk”) have no significant financial effect, and greenwashing, measured as the gap between talk and walk, also harms financial outcomes. This suggests that while impression management may yield immediate reputational dividends, it remains a high-stakes strategy that carries significant long-term credibility and valuation risks if discrepancies between the narrative and operational reality are eventually exposed. In our study, impression management theory will help to capture whether environmental content is foregrounded or buried within the CEO letter and demonstrate if sustainability is presented as strategic priority or as a compliance afterthought.

2.3.2 Grice's Cooperative Principle and Paltering

The literature approaches greenwashing not only as a moral failure but also as a pragmatic manipulation of language (Lyon & Montgomery, 2015, p. 223). One of the most major analyses called Gricean Cooperative Principle positions effective communication on four foundational maxims: Quality (truth), Quantity (sufficiency), Relation (relevance), and Manner (clarity), as described in Table 1. From the corporate social responsibility perspective, greenwashing occurs when there is a systematic “flouting” of these maxims (Lyon & Montgomery, 2015, p. 224).

Table 1: Maxims of the Gricean Cooperative Principle

Maxim	Linguistic Expectation	Strategic Violation (Greenwashing)	Resulting Deception
Quality	Provide truthful, evidenced data.	Paltering: Leveraging literal truths to imply a false overarching narrative.	The audience infers a "green" identity based on a single, isolated factual claim.
Quantity	Provide a sufficient informative contribution.	Selective Omission: Withholding "unfavorable" data points regarding ecological degradation.	The report becomes "functionally deceptive" by providing an incomplete data set.
Relation	Maintain relevance to the subject.	Red Herring: Diverting scrutiny from core systemic harm toward minor, positive CSR initiatives.	The focus shifts from high-impact pollution to low-impact philanthropic gestures.
Manner	Avoid ambiguity and achieve clarity.	Obfuscation: Utilizing technical jargon to mask simple environmental failures.	"Retrospective rationalization" makes harm appear as a complex, unavoidable byproduct.

Communicators find ways to cross these linguistic boundaries to achieve their goals instead of being direct and communicating the true message. The main mechanism for making deception is a framework called “paltering”. This strategy involves being technically honest, but fundamentally dishonest. Paltering is particularly different than lying outright since entities maintain the technical truth. Buller & Burgoon (1996, pp. 203-205) observed that the reader evaluates the messages based on perceived completeness and relevance. Consequently, when the message giver withholds unfavorable information, they violate the quantity and relevance principles of communication.

Recent research has shown that bending the truth is common and it is particularly through selective disclosure and omission in corporate communication. These strategies create a cycle of retrospective rationalization. Companies continue their operations in their best interest and use these linguistic tricks to avoid getting criticized or creating a favorable narrative to justify their negative impact. This cycle of manipulating conversational rules enables firms to pretend to meet green demands of stakeholder meanwhile continuing the industrial impact behind confusing, yet technically accurate words (Gorovaia & Makrominas, 2025, p. 428-430; Lyon & Montgomery, 2015, p. 223-224). The Gricean violations are directly operationalized in the analysis of the CEO letters, capturing violations of the Quantity maxim through an excess of future-oriented language as well as the violations of the Manner maxim by the use of vague or distant commitments that obscure verifiable accountability.

2.4 Linguistic Frameworks

2.4.1 Systemic Functional Linguistics

Michael Halliday's Systemic Functional Linguistics (SFL) moves beyond treating language as a static set of grammatical rules. The SFL model views the language as a "social semiotic" where language is seen as an organized resource for making meaning within a specific context (Halliday, 1978). This view promotes that grammar is a system of choices where every linguistic decision is chosen intentionally. The speaker or the writer makes this linguistic decision based on the goals and the relation to the environment.

SFL is designed with the foundation that every instance of language fulfills three overarching metafunctions: ideational, interpersonal, and textual (Halliday, 1978). These metafunctions provide multidimensional lenses through way to analyze organizational discourse by mapping the connection of internal structure of the language and the external social world. Interpersonal metafunction is especially important to this study since it shows how CEOs use language choices rather than just factual reporting to position their companies in relation to stakeholders and build sustainability commitment.

One of the most critical SFL mechanisms is nominalization as a concept of grammatical metaphor (Halliday, 1985). This occurs when the writer or speaker deviates from being "congruent" (direct) to a metaphorical, abstract tone. In business context, the concept of grammatical metaphor is mostly common in the form of business nominalization. The usual way of language involves the use of verbs to show who is doing the action. However, corporate speeches treat human decisions and processes as if these were solid and unchangeable objects. This process is called reification, and organizations apply this by turning verbs into nouns.

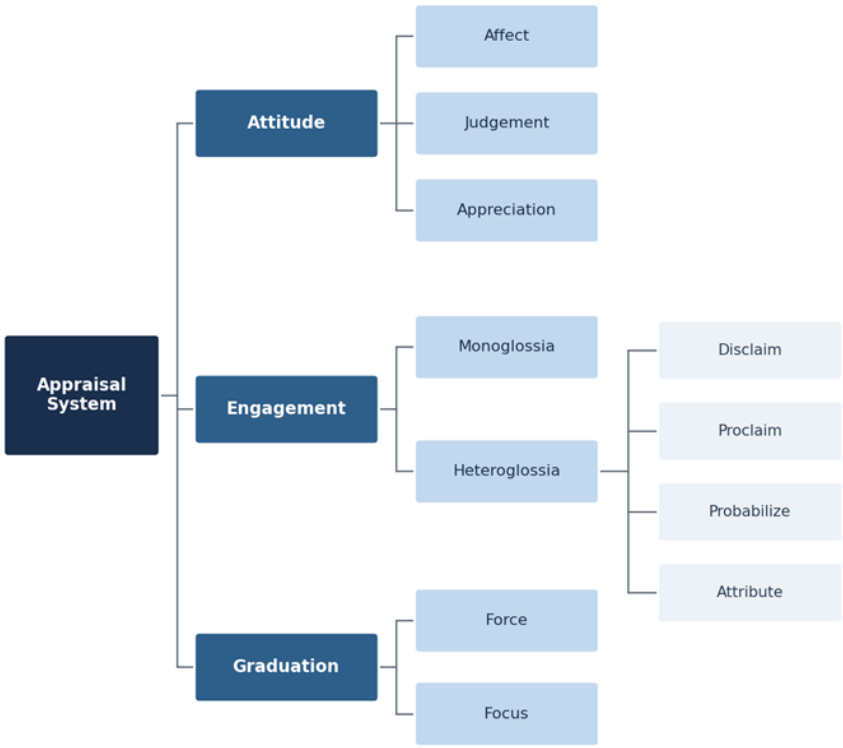
Corporate rhetoric explains that nominalization serves as a function for distancing (Halliday, 1985). For instance, corporate organizations remove the active agent and the affected subject when a process with a verb "we laid off two thousand workers" is transformed into a noun "the workforce reduction". Hence, reification allows organizations to present controversial or painful events as a fixed, inevitable fact. As a result, the noun version shields the decision makers from direct accountability and frames outcomes as abstract phenomena. In this study, high nominalization density in environmental sentences is treated as a linguistic signal of accountability avoidance, consistent with the impression management strategies documented by Brennan and Merkl-Davies (2013, pp. 110-112). SFL provides the theoretical foundation for understanding how the language structure of the CEO letters can function as a signal of accountability avoidance.

2.4.2 Appraisal Framework

Martin and White (2005) developed Appraisal Framework that offers a more specialized way to analyze interpersonal metafunction. The framework focuses on the way speakers or writers express their emotions, make judgments about human behavior, and evaluate the quality of objects. It explains the linguistics resources used to convey attitudes, align or distance oneself from other viewpoints, and adjusts the strength of evaluative statements. In other words, appraisal describes how language encodes evaluation. This is particularly useful for analyzing corporate communication because it reveals how organizations construct credibility, express commitment, and manage stakeholder expectations through evaluative language. The framework is categorized into three subsystems as follows: Attitude, Engagement, and Graduation. Although we excluded the Appraisal based variables (C5 and C6) from the

operational GWRI for reproducibility reasons, the framework still contextualizes why the four rule-based variables capture meaningful variation. Firms that lean on inflated, one-sided or emotionally amplified environmental language tend to be the same firms whose CEO letters score poorly on claim specificity, temporal accountability, and structural emphasis.

Figure 1: Appraisal System (Martin & White, 2005)



The Attitude system provides a structural way to analyze how organizations express feelings, values, and evaluations through Affect, Judgement, and Appreciation (Martin & White, 2005). For example, in this study, high Attitude in environmental sentences, where business asses their own sustainability performance in highly positive terms, indicate that language may be acting more promotional than informational, which is a fundamental feature of greenwashing (Sneideriene & Legenzova, 2025, p. 3). Furthermore, the Engagement approach makes a distinction between heteroglossic statements that take into account different points of view and monoglossic statements that are presented as unquestionable, absolute facts. This study treats a high percentage of monoglossic environmental phrases in CEO letters as a rhetorical predictor of the danger of greenwashing, which implies the company is claiming sustainability credentials rather than reporting them. Finally, the Graduation system addresses how writers use Force and Focus to scale meaning up or down. This study treats systematic upward graduation in environmental language, where intensity is amplified beyond what disclosed facts warrant through expressions like "deeply committed" or "truly sustainable", as a linguistic marker of the impression management gap where tone consistently surpasses substance (Martin & White, 2005; Sneideriene & Legenzova, 2025, p. 3).

2.5 ESG Performance and Stock Market Performance

2.5.1 Mechanisms Linking ESG to Financial Markets

Understanding the economic relevance of ESG metrics in capital markets is essential to making a connection between environmental sustainability efforts and financial performance of the underlying asset. If investors do not incorporate ESG factors into valuation, sustainability performance may primarily influence short-term market sentiment rather than contribute to long-term valuation (Khan et al., 2024, p. 20). The empirical finance literature identifies several conditional mechanisms. First, the literature shows that ESG-related information can influence capital markets (Cao & Zeng, 2025, p. 721). Additionally, ESG metrics can provide downside-risk protection by reducing exposure to specific risks, attracting investors who prioritize sustainability. This enhances resilience during crises through increased trust and social capital and amplifies the effects of sustainability narratives in cases where measuring ESG metrics precisely is difficult (Kansoy & Stasiulaitis, 2025, p. 4). These mechanisms do not always guarantee outperformance but instead considering that it heavily influences expected cash flows, discount rates, or even perceived tail risk is important. Therefore, it shapes asset prices in situations and contexts where sustainability considerations are significant (Krüger, 2015, p. 305).

Previous empirical studies have shown that markets show stronger reaction to news that are negatively related to ESG than positive news or developments. According to research, negative corporate social responsibility (CSR) events are reacted with strong negative abnormal returns meanwhile positive news results with insignificant or weak market responses (Krüger, 2015, pp. 304-305). This asymmetry in the reaction indicates that ESG strength functions as a downside protection for withstanding negative shocks better. For this reason, companies with strong sustainability practices perform resilience in stressed markets, which is a result of strong stakeholder trust (Lins et al., 2017, pp. 1785-1813).

Outstanding ESG practices extend beyond reputational considerations by reducing certain risks of exposure and making the performance more stable (Berk et al., 2023, pp. 146-147). For instance, carbon intense companies have more risk as they are more sensitive to changes in environmental regulations and carbon taxes. Changes in the regulations or taxes can significantly affect expected future cash flows. If a high sustainability risk company misrepresents its progress in carbon reduction, the market can punish with the valuations due to anticipated fines, reputational damage, legal liabilities, or loss of trust (Walker & Wan, 2012, pp. 234-238; Bernini & La Rosa, 2024, p. 432). Volkswagen's Dieselgate scandal is one of the most famous examples on how negative ESG news can rapidly decrease market valuation. Regulators revealed that Volkswagen cheated on emission tests in millions of diesel vehicles with manipulative software in September 2015. Investor's reaction after the disclosure caused one third of the stock price decline within two days after the scandal became public (Bachmann et al., 2023, pp. 489-491).

2.5.2 ESG Materiality, Measurement Challenges, and Rating Divergence

The impact of ESG issues varies depending on the industry as the effect cannot be generalized due to the variations on results. For this reason, financially material ESG factors should be distinguished based on the relevance for the specific industry as more relevant factors create more economically significant outcomes. Previous research demonstrates that financially material ESG factors for the given industry such as carbon emissions for energy intensive sectors or data privacy in technology firms are more linked to the financial performance of the

company than the immaterial sustainability indicators (Khan et al., 2016, p. 1698). Thus, investors should focus on relevant ESG dimensions rather than a uniform concept. Furthermore, distinguishing between sustainability issues that are central to the company's operations and the ones that are less significant is also helpful in explaining the different ways and consequences that greenwashing can occur. According to Walker & Wan, investors and stakeholders are concerned with the alignment of environmental claims and actual practices as the legitimacy impacts future access to resources and long-term performance (Walker & Wan, 2012, p. 228). This underscores how greenwashing is a fragile attempt to obtain legitimacy and highlights the misstating risk.

Another fundamental mechanism that ESG information influences financial performance is due to demand-based pricing by investor attention and preferences. Morningstar Sustainability Ratings demonstrates that investors relocate their capital towards highly rated funds from weaker scored fund withdrawals in cases where sustainability information is easily interpretable and visible. Notably, the rating does not provide additional information about firms but presents the already available ESG data in a simpler, easier to understand format. Consequently, the flow of the capital appears to be driven by visibility instead of changes in the firms' fundamentals (Hartzmark & Sussman, 2019, pp. 2791-2792). This visibility mechanism can be relevant for corporate disclosures such as CEO letters as they function as attention shaping tools. Potentially, CEO letters can influence investor perceptions before more performance metrics become available.

Evaluating ESG-related financial returns has complications due to measurement challenges. Currently, a universally accepted ESG performance metric does not exist. For this reason, rating agencies often provide different assessment scores for the same firm. Previous empirical evidence demonstrates that there is significant disagreement in sustainability performance evaluation and correlations between ESG ratings is imperfect. The divergence of ratings is explained primarily by differences in scope, measurement choices modeling of the indicator instead of weighting discrepancies (Berg et al., 2022, pp. 1316-1320). Hence, consensus absence creates informational noise in ESG signals and overcomplicates the interpretation of sustainability performance for investors.

2.5.3 Narrative Disclosures, Institutional Context, and Market Efficiency

Since ESG measurements are imperfect, corporate communications can become a significant supplementary signal for investors if they attempt to evaluate a firm's sustainability performance. Investors may rely on narrative disclosures with qualitative descriptions of sustainability initiatives in situations where rating agencies disagree, or the ESG metrics are difficult to comprehend (Christensen et al., 2022, p 148). Therefore, managerial communication can shape perceptions of ESG performance depending on the way they want to promote themselves (Bendig et al., 2024, p. 1). On the other hand, firms may behave opportunistic and strategically highlight sustainability achievements in ways that is not in line with the performance indicators (Gonçalves et al., 2020, pp. 417-427). Nevertheless, empirical evidence suggests that ESG performance and financial outcomes are inconclusive. An analysis of over 2,000 studies indicates that findings on the ESG-financial performance relationship are often mixed and depend on factors such as methodology and context, although many studies report non-negative associations (Friede et al., 2015, p. 211). The variation denotes that the existence of multiple mechanisms of ESG influences financial outcomes. A recent study of asset-pricing underscored that investors may find dynamic changes in ESG signals more relevant than static ESG levels. Studies of ESG momentum and ESG volatility showed that

expected returns is influenced when firms make improvements in sustainability performances relative to other similar firms and instability in ESG scores can be interpreted risky (Berk et al., 2023, pp. 135-159). Thus, the investors not only consider the level of ESG but also the trend of it through the lenses of a market pricing perspective.

Additionally, institutional context is also significant to evaluate how ESG signals are interpreted by investors. Prior evidence from Nordic markets showed that consideration of sustainability is embedded in reporting norms, corporate practices which makes the Nordic region a relevant market to study ESG related effects (Vaihekoski & Yahya, 2023, pp. 164-189). Credibility of ESG disclosure can be higher in markets where there is a high level of transparency and established standard reporting practices. Consequently, high transparency might also raise stakeholder expectations regarding sustainability commitments of the company.

Overall, previous literatures show that information regarding ESG can influence financial markets with several different mechanisms like investors demands for sustainability, disagreements in measurements, and evolving perceptions about firms' sustainability trajectories. These influences imply that narrative disclosures like CEO letters may have an influence in shaping the investors' behavior.

A foundational constraint on the mechanisms described above is widely known as the Efficient Market Hypothesis (EMH), originally developed by Fama (1970, pp. 383-384). In its semi-strong form, the EMH posits that all publicly available information is already incorporated into asset prices, implying that no publicly available signal, including sustainability narratives in CEO letters, should systematically predict future returns. However, this assumption relies on the premise that information is costless and efficiently processed by all market participants. Grossman and Stiglitz (1980, pp. 393-394) challenge this view by arguing that if information is costly to acquire and interpret, markets cannot be perfectly efficient. In this context, ESG-related disclosures, particularly qualitative and narrative forms such as CEO letters, may not be immediately or fully incorporated into prices. This creates room for investor heterogeneity in attention and interpretation, allowing such disclosures to influence asset prices, at least temporarily. Therefore, narrative ESG communication can be relevant in shaping market outcomes, especially when information is complex, ambiguous, or difficult to quantify.

2.6 Stock Market Reaction to Greenwashing Events

Empirical research that focuses on greenwashing events have a different approach by treating greenwashing as a negative shock for credibility and tests the negative reaction by evaluating exposures, allegations and regulatory actions (Dorfleitner et al., 2025, p. 1). Unlike cross-sectional studies that examine the general relationship between sustainability disclosure and financial performance, event-study designs isolate the immediate price response to a discrete, publicly disclosed greenwashing incident, providing a more direct test of whether markets penalize deceptive environmental communication. A conducted study of 296 greenwashing events within STOXX Europe 600 firms showed that average market responses are not uniform. The study results showed that abnormal negative returns are more observable when the firms are smaller, when the case is financially material (where consequences for investors are more extreme) and when there is a compliance related allegations that harm ESG reputation (Dorfleitner et al., 2025, p. 8). Xu et al, (2025, p. 15) had complemented evidence with an analysis of 121 global greenwashing news starting from 2015 Paris agreement. The study

results demonstrated that the stock market reacts more negatively when the news includes more concrete evidence and when the underlying firm previously had strong ESG performance. On the other hand, there is another report that was conducted in 2024 that suggests markets do not consistently penalize greenwashing as no statistically significant CARs were found in greenwashing announcements in Europe (Teti et al., 2024, p. 8). Taken together, these studies indicate that market reactions to greenwashing are conditional rather than automatic: they tend to materialize when the alleged misconduct is verifiable, financially consequential, or represents a sharp contrast with a firm's prior sustainability reputation.

2.7 Hypothesis formulation

Despite the expanding body of research on sustainability reporting and greenwashing, the existence of greenwashing risk in CEO letters and its connection to external ESG rating and financial market outcomes have not been thoroughly studied. Taking into consideration this gap and the literature presented above, this study will examine, as previously mentioned, whether CEO's sustainability communications accurately reflect the actual sustainability performance or whether they may contain features of greenwashing. Furthermore, the study aims to investigate how the degree of greenwashing risk affects the subsequent stock market performance within B2B Nordic companies. In order to achieve the aim of the study, two hypotheses have been formulated.

The first hypothesis expands upon the idea of signaling theory and impression management. In information asymmetry, a signal needs to be both expensive and verifiable in order to be considered credible (Connelly et al., 2011, pp. 39-41). CEO letters, however, frequently rely on ambiguous or exaggerated sustainability rhetoric, low-quality signals that hardly match the quantitative measurements employed by ESG agencies. ESG evaluations serve as a check on managerial discretion because they are based on external data and specified standards rather than corporate narratives (Zhang et al., 2025, p. 9). Therefore, there should be a noticeable discrepancy in the form of lower third-party ESG scores for companies who utilize their CEO letters for greenwashing, which is typified by low specificity or purely aspirational language.

Building on this, our second hypothesis explores how the capital markets price these discrepancies. While the existing literature is somewhat divided, a coherent narrative has begun to emerge: "symbolic" green talk that lacks substantive action eventually erodes financial performance (Walker & Wan, 2012 p. 227). As Ghitti et al. (2024, p. 907) point out, firm value does not always decline immediately; rather, it tends to erode gradually as the gap between corporate rhetoric and reality becomes too wide for investors to ignore. This is especially true in the Nordic region, where strong institutional frameworks and stakeholder-oriented governance structures shape corporate practices (Gjølberg, 2010, pp. 208-209). Beyond just the stock market, there is a practical B2B risk to consider: if business partners perceive a firm's sustainability claims as opportunistic, they may pull back from long-term relationships, causing real-world damage to revenue and stability (Walker & Wan, 2012, p. 231). Taken together, these factors suggest that CEO letters signaling a high risk of greenwashing are a reliable lead indicator for weaker stock performance in the year following their publication.

We note that the empirical evidence on whether equity markets systematically penalize greenwashing remains mixed, particularly in European contexts where some studies find no significant abnormal returns following greenwashing events (Dorfleitner et al., 2025, p. 8). In the Nordic setting, where sustainability expectations are deeply embedded in institutional

norms, market reactions to incremental narrative-level signals may be more muted than reactions to more explicit greenwashing scandals. H2 therefore represents a directional expectation based on theoretical reasoning, while allowing for the possibility that narrative-level greenwashing risk may not produce detectable short-term return effects in this context.

H1: Firms with higher greenwashing risk in their CEO letters receive lower third-party ESG ratings.

H2: Firms with higher greenwashing risk in their CEO letters experience lower stock returns in the subsequent year.

3. Methodology

This chapter outlines our methodology including how the research philosophy, approach and design. It describes how data was collected and analyzed using the Greenwashing Warning Risk Index and Regression Analysis. Furthermore, it discusses the validity, reliability, and some limitations faced during the process.

3.1 Topic Selection

As students in finance, we have a strong academic background in financial analysis and investments. However, when selecting the research topic, we did not seek to focus on purely financial research; instead, we searched for a topic that combined our academic specialization with personal interest and aimed to focus on something that reflects current financial and societal challenges. Ultimately, we have concluded focusing on finance and sustainability. Our interest in sustainability has grown because of numerous ESG-related academic projects and independent research, which have made us more conscious of the expanding role that sustainability plays in risk assessment, corporate valuation, and investment decisions.

We have noticed that investors are depending on executive communication and corporate ESG disclosures, especially in regions like the Nordics where sustainability standards are strong and companies market themselves as responsible in order to meet stakeholders' expectations. This sparked debate on the reliability of sustainability narratives and the degree to which they accurately represent their performance. In addition, we were curious about the idea of "greenwashing" and the potential impact of discretionary narrative disclosures, like CEO letters, on investor views and subsequent market performance.

Even though sustainability and ESG reporting are extensively researched, we identified that, particularly in a Nordic B2B setting, there is relatively little empirical research relating executive sustainability rhetoric to third-party ESG ratings and stock market performance. This fact served as a motive to choose this research area. In addition to helping investors and businesses navigate the increasing complexity of ESG-related information, we hope that this research will further knowledge of the relationship between sustainability communication and financial markets.

3.2 Research Philosophy

The methodological decisions that are presented in the sections that follow are not independent of one another, as each choice was informed and constrained by the one that preceded it. A research philosophy, commonly known as the research paradigm, is considered a foundation for how a research study is designed, and it includes a collection of fundamental presumptions and convictions concerning the development of understandings. These assumptions are essential to consider for researchers as it gives them a framework for comprehending the nature of reality, how knowledge is created, and the approaches and strategies utilized to investigate and respond to research problems (Kivunja & Kuyini, 2017, pp. 26-28). According to Junjie and Yingxin (2022), there are two major types of research paradigms that are positivism and interpretivism.

Positivism places a strong focus on the objective and empirical examination of reality in order to identify patterns through methodical observations and measurements. This viewpoint

emphasized minimizing bias, where knowledge is derived from observable data rather than subjective interpretations and assumptions (Saunders et al., 2019, p. 136). The positivism viewpoint is primarily associated with quantitative methods, such as statistical analysis, and surveys, which enable the examination of relationship between variables (Omodan, 2022, p. 277). Interpretivism on the other hand is often linked to qualitative or mixed research methods, and especially appropriate when a study subject needs a deeper understanding and when the body of knowledge is restricted. By concentrating on the depth and diversity of individual experiences rather than just quantifiable results, research implement this paradigm seek to identify the underlying meanings, symbols and interactions that influence human behavior (Saunders et al., 2019, p. 143; Gunbayi, 2020). The most used assumptions in scientific research include ontology, epistemology, and axiology (Kivunja & Kuyini, 2017, pp. 26-28). These components of paradigm will be presented and discussed in the subsections below.

3.2.1 Ontology

According to Kivunja and Kuyini (2017, p. 27), the focus of the philosophical field of ontology are the assumptions researchers make in order to think something makes sense or is real, or the fundamental characteristics of the social reality they are investigating. Ontology is essential because it helps to get a deeper understanding of how the researcher makes meaning of the data they collect. Within a positivist paradigm, reality is viewed as objective, unchanging and uneffaced by human perspectives, and is measurable similar to natural sciences, where everyone has the same sense of reality. Conversely, interpretivists believe that reality is subjective, asserting that reality is socially formed and shaped by individual meanings, connections, and contextual understanding.

The ontological stance of this study aligns with an objectivist paradigm within positivism as the research assumes that greenwashing risk, ESG ratings and stock market performance can be measured as observable, structured aspects of reality. ESG ratings are regarded as standardized external evaluations, stock performance is based on quantifiable market data, and greenwashing risk is operationalized through predetermined textual indicators, all of which assume that the phenomena being studied have stable characteristics that can be empirically captured. Even though individuals may interpret greenwashing differently in practice, this study does not examine perceptions and takes an objectivist, positivist ontological perspective by treating greenwashing risk, ESG ratings and stock market performance as measurable.

3.2.2 Epistemology

Epistemology focuses on the fundamentals of knowledge, including its nature, forms, and acquisition and communication with other people. It emphasizes the nature of human understandings and knowledge that the researcher can potentially obtain to expand, develop, and broaden knowledge in the subject of study (Kivunja & Kuyini, 2017, p. 27). This philosophy deals with the relationship between the researcher and what can be considered as valid knowledge. Every paradigm holds a district viewpoint on what defines valid knowledge. From a positivistic perspective, epistemology, knowledge is considered valid only if it is observable and measurable, maintaining an objective stance. On the other hand, an interpretivist perspective suggests that knowledge is subjective and linked with emotions and values. The goal of this philosophy is to reduce the gap between the researcher and the subject of the study.

Similarly to ontological paradigm, we believe that this study adopts a positivistic epistemological paradigm, as it is grounded in the assumption that valid knowledge is

generated through objective measurement, systematic observation, and empirical testing. In this study, "greenwashing risk" is not treated as a subjective feeling, and we do not investigate *why* CEOs wrote what they did and how they feel about it, or how investors perceive greenwashing risk. Instead, "greenwashing risk" is treated as a quantifiable variable derived from predefined linguistic and thematic indicators within CEO letters. In addition, by using third party ESG ratings, and stock market performance, we ensure that the data remains independent of the personal bias, fulfilling the positivistic epistemological stance.

3.2.3 Axiology

Axiology refers to the role of values in the research process, specifically the extent to which a researcher's personal beliefs, biases, and value judgements may influence the study's outcomes. As noted by Kivunja and Kuyini (2017, p. 28), axiology also encompasses the ethical considerations that must be addressed when designing a research proposal, including how researchers identify, evaluate and understand notions of appropriate and inappropriate behavior in relation to the study's aims. Positivists believe the research process to be value-free. This means that the researcher tries to eradicate personal bias and remain independent during the process. This supports the positivist stance that reality is objective and measurable without interference from the researcher's personal perspective. As a result, values are regarded as something that should be regulated, bracketed or eliminated in order to guarantee the validity and repeatability of findings. To the contrary, interpretivist believe that values are intrinsic and closely linked to research process because the researcher actively participates meaning interpretations, incorporating their own experiences, background and belief into the processed of creating knowledge (Kivunja & Kuyini, 2017, pp. 28). Yilmaz (2013, p. 316), also notes that interpretivist paradigm openly acknowledges that research is value-laden and bias it presents.

This study adopts a value-free axiological stance. In line with this position, we, for example, do not define or interpret greenwashing from our own perspective, instead, we rely fully on theoretical concepts and predefined measurement criteria to ensure consistency and neutrality throughout the analysis of CEO letters. By using external ratings and stock performance, the study maintains a distance between our own values, beliefs and interpretations and the phenomena being measured.

3.3 Research Approach

In an empirical study, establishing an appropriate methodological approach is essential and can be helpful in aligning the research with the aim, data, and analysis. In research methodology, two primary approaches are commonly distinguished: deductive and inductive. A deductive approach is mostly associated with quantitative research and begins with a well-established idea, concepts, and hypothesis, which later investigates their applicability in particular empirical situations. A researcher develops an established hypothesis based on theoretical assumptions. These are then empirically evaluated through data collection and statistical analysis in order to either validate or refute the original hypothesis. For this reason, the deductive approach is referred to as moving from general to particular (Saunders et al, 2019, p. 145-155).

An inductive research approach, on the other hand, refers to the study that is emergent and exploratory in nature, where theory is developed from the observation of empirical reality Daniel & George, 2025, pp. 3-4). Glaser and Strauss (1967) state that this method is particularly useful for investigating obscure occurrences, complex societal circumstances, or personal

experiences that are difficult to quantify. In contrast to the deductive approach, the inductive approach is referred to moving from particular to general, since it entails transitioning from individual observations to general pattern description. This approach is particularly suitable for qualitative research and for a subjective perspective (Saunders et al, 2019, p. 145-155).

However, Bamberger (2018, p. 2) discussed a third research approach: abductive reasons, which refers to a way of developing plausible explanations for unexpected findings. Abductive reasoning is an approach where the researcher moves back and forth between data and theory in an iterative way to develop the most likely interpretations of a phenomenon. It begins with an observation that cannot be adequately explained by current theory. In contrast to a deductive and inductive approach, this method entails creative inference using both theoretical concepts and empirical insights to improve knowledge (Dellsén, 2024; Bamberger, 2018, p. 2). Abductive reasoning can be described as a mix of inductive and deductive approaches.

In our study, we have implemented a deductive approach. An inductive and abductive reasoning are not sufficient because the study does not aim to develop or iteratively refine theory from the data, therefore, we believe that the deductive approach is the most suitable approach to accomplish the research objective. A deductive reasoning is used in this study, as the analysis begins with establishes theoretical assumptions and hypothesis regarding greenwashing, ESG assessments, and financial markets performance. Based on previous research, the paper formulates expectations about how greenwashing risk in CEO letters should relate to third party ESG ratings, and subsequent stock performance. These expectations are then empirically tested using quantitative textual measures, and market data. Therefore, the hypothesis testing nature of the research design aligns with a deductive logic of inquiry.

3.4 Research Design

3.4.1 Research Strategy

To situate a study within an appropriate methodological framework, it is essential to classify the study according to its purpose. The four major purposes include exploratory, descriptive, explanatory, and predictive. Exploratory research is usually conducted when there is no or little information about a specific phenomenon or subject. The aim of the descriptive research is to describe the detailed characteristics of that phenomenon. Explanatory research aims to explain the cause and effect of the phenomenon being studied. Finally, predictive research attempts to generalize findings and make predictions about future outcomes based on established hypotheses or models (Saunders, 2019, pp. 174-185).

In order to address the topic of greenwashing in the Nordic B2B sector, this study adopts a multi-layered research strategy that is primarily explanatory, while also incorporating exploratory and descriptive parts. The core of the research is explanatory as it seeks to move beyond mere observation to identify the correlations and causal mechanisms, between CEO language and external benchmarks, specifically looking into at the third party ESG ratings and stock market performance. This choice is further supported by the need to bridge the gap in current studies, for example Hummel & Schlick (2016) and Kim & Lyon (2015), have established foundational links between disclosures quality and firm value, but there remains a need of evidence regarding these dynamics within Nordic B2B firms. To account for this, the study utilizes an exploratory component to categorize unique linguistic patterns of greenwashing in this specific geographic and industrial context. Moreover, the descriptive

strategy is incorporated to measure the amount of greenwashing risk using Natural Language Processing (NLP) analysis and provide an overview of the ESG ratings and stock performance. By integrating all these strategies, the study offers a thorough account of greenwashing risks and converting them into market results.

3.4.2 Methodological Choice

A research method can be categorized as either qualitative or quantitative. Qualitative research is a research method that focuses on understanding individuals' experiences, perspectives and behaviors in their natural environment. The method is often used to investigate complex phenomena that are difficult to quantify, such as beliefs, attitudes, and emotions. Data for qualitative research is often collected through methods such as observations, interviews, and focus groups. The information gathered is non-numerical and may consist of text, audio, and images (Yilmaz, 2013, p, 311). According to Yilmaz (2013, p. 315), one of the defining characteristics of qualitative research is its emphasis on context and the subjective interpretation of data. Rather than attempting to generalize the results to a broader population, qualitative researchers strive to understand the meaning and relevance of the collected data by evaluating it within its context. Quantitative research, on the other hand, focuses on systematic collection analysis on numerical data. This approach is often used to investigate correlations between variables and to make predictions or generalizations about a broader population based on a sample. Quantitative research often involves collecting data using methods such as surveys, experiments, and structured observations, and then evaluating data using statistical techniques. One of the defining characteristics of quantitative research is its emphasis on objectivity and the use of standardized measurements. Quantitative researchers use rigorous methods to collect and analyze information in order to minimize the effects of personal bias and subjectivity (Yilmaz, 2013, p 311).

As we do not investigate subjective experiences or interpretations of CEO letters, the most suitable research approach will be a quantitative approach. The research questions focus on identifying measurable patterns and statistical relationships between predefined variables 1) textual indicators of greenwashing risk, 2) third party ESG ratings, and 3) financial market outcomes. Because all three variables can be transformed into numerical variables, a quantitative approach will enable a systematic comparison, hypothesis testing, and generalizable conclusions. Quantitative methods are particularly appropriate in this study because we will be using observable data rather than exploring subjective meanings. In this study, greenwashing is not interpreted from a personal viewpoint, instead, it is operationalized through objective indicators extracted via Natural Language Processing (NLP), Similarly, ESG ratings and stock performance are inherently numerical, making statistical analysis the natural choice for assessing correlations associations.

3.4.3 Sampling Method

The sample consists of publicly listed B2B companies from the Nordic countries including Sweden, Norway, Finland and Denmark. The initial plan was to focus solely on Sweden listed firms. However, during the preliminary data screening, we realized that the amount of Swedish B2B firms publishing CEO letters was insufficient and not enough to produce a generalized study. Therefore, to ensure generalized and avoid a sample that was too narrow, we expanded the scope by including remaining Nordic countries. We later recognized that this decision not only solved the problem of sample size but will strengthen our study. The Nordic countries share highly similar sustainability regulations, and reporting culture. Moreover, Nordic investors are known for their strong expectations regarding corporate transparency, making the

whole region particularly relevant for examining greenwashing and sustainability communication (Strand et al., 2015, pp. 1-13; Vaihekoski & Yahya, 2023, p. 166). Therefore, including all four countries, therefore strengthens the comparability and external validity of the findings.

The chosen industry for this study is B2B firms, rather than all listed companies, due to several reasons. Firstly, B2B firms often operate in industries with heavy environmental footprints (e.g., manufacturing, industrial, and energy-related services). Accordingly, in this industry, greenwashing is increasingly noticeable. Secondly, unlike B2C firms, B2B relies on long term stakeholder relationships, affecting the way sustainability communication is designed. Thirdly, focusing on B2B firms is relevant from a financial perspective. Unlike B2C firms, B2B companies depend heavily on long-term contracts, supply-chain relationships, and inter-organizational trust to sustain their operations and resource base (La Rocca et al., 2019, pp. 150-151). These structural characteristics make B2B firms particularly sensitive to reputational and environmental risks, as any disruption to supplier or client relationships can directly affect revenue stability and overall business value. Consequently, investors in B2B firms are more likely to scrutinize sustainability performance, since environmental hazards and greenwashing risks have the potential to undermine the long-term contracts and supply-chain stability that these firms depend on.

The selected time frame for the sample is 2022 to 2024. We believe that three-year period of appropriate for this study, because, firstly, it captures the recent and relevant window to observe changes in stock performance. Secondly, a three-year horizon balances temporal depth with feasibility, since a longer period would significantly increase the volume of textual data to process, affecting the quality of the study. Finally, the year 2024 is the most recent year for which complete information is available for all firms in the dataset. Data for 2025 has not been fully published for all firms, and we did not want to risk inconsistencies and missing values.

Determining the appropriate sample size is often shaped by practical constraints such as available time and resources (Saunders, 2019, pp. 174-185). In relation to this study, resource constraints were not a limiting factor, but the time required may have influenced the sample. Collecting, sorting and analyzing large amounts of data, both textual and numerical, demands an enormous amount of statistical and manual work. Therefore, the sample was created to be both manageable within the thesis timeline and large enough to provide a statistical analysis. Our initial goal was to analyze approximately 200 firms, 50 firms from each country. The firm identification process was carried out using the LSEG Refinitiv system, where firms were sorted by market capitalization from largest to smallest for each country separately. This top-down approach was chosen because larger firms are more likely to have complete ESG datasets available due to stricter reporting requirements and greater investor scrutiny. Once the ranked firm list was generated, all firms were manually reviewed to verify that only B2B firms were included in the sample. Firms that were primarily B2C in their business model were excluded at this stage. Subsequently, firms with missing ESG data or unavailable CEO letters for any of the three sample years were removed to ensure a balanced and complete dataset. This process resulted in a final sample of 113 firms - 42 from Sweden, 21 from Norway, and 26 and 24 from Finland and Denmark respectively.

3.4.4 Collection of Data

A common distinction in research methodology is between primary and secondary data. Primary data refers to the information that the researcher collects for the first time, while

secondary data is the information that has been collected and created by others (Ajayi, 2015, p. 1). As noted by Ajayi (2025, p. 1), the main difference between these two is that secondary data is merely the analysis and interpretation of primary data, whereas primary data is factual and unique. Since this study aims to analyze information that is already publicly available, such as CEO letter, ESG ratings and historical stock performance, it will exclusively rely on secondary data.

3.4.4.1 ESG Ratings

After finalizing the sample of firms, ESG data were gathered using the LSEG Workspace Excel add-in. This tool enabled efficient extraction of ESG-related variables for all companies in the sample at once. By leveraging its built-in formula functions, ESG ratings were retrieved simultaneously for all firms, eliminating the need for individual data collection. This approach greatly reduced the time required and enhanced the consistency of the dataset. The ESG variables collected included the overall ESG rating, environmental score, social score, governance score, and controversial ESG score. These metrics offer a comprehensive overview of firms' sustainability performance across multiple dimensions. During data collection, some firms were excluded due to missing ESG data in the LSEG Workspace. These cases were removed to maintain a complete and consistent dataset across all variables.

3.4.4.2 CEO Letters

For the CEO letters, all annual reports were manually downloaded for each company for the years 2022, 2023, and 2024. This ensured that the analysis was based on original and official company disclosures. After collecting the reports, the CEO letter sections were manually extracted from each document. Only the relevant text sections were included, and all other parts of the reports were excluded from the dataset. The extracted CEO letters were then transferred into an Excel file, where each observation was linked to the corresponding firm and year. The text extraction process was carried out manually to ensure accuracy and consistency in identifying the correct sections across different report formats. Once the dataset of CEO letters was completed, it was prepared for further analysis

3.4.4.3 Stock Performance

For stock prices, the data collection process followed a similar approach to the one used for ESG ratings. LSEG Workspace was used through its Excel add-in to extract the relevant financial data for all firms in the sample simultaneously. This was done by inputting each company's ticker (RIC code) and selecting the required variable, namely the closing price. Stock prices are collected for 2022, 2023, 2024, and 2025 in order to ensure a correct time alignment between CEO letters and subsequent market reactions.

3.5 Greenwashing Risk Measurement: Overview and Design Philosophy

Before describing what Greenwashing Warning Risk Index (GWRI) is, it is worth understanding why existing measurement tools are insufficient to the scope of this study. There are several measurement tools that measures tone, sentiments, and linguistic risk in corporate disclosures, however, none of them, individually or in combination, can operationalize the specific theoretical constructs this research requires to test its two hypotheses. Although tools like VADER and the Loughran-McDonald (2011) lexicon are very useful for gauging broad sentiment, they are unable to differentiate between "honest" positivity and the calculated, compensatory rhetoric employed in greenwashing. In a similar vein, specialist models such as Gorovaia and Makrominas (2025) were validated on CSR reports, which are more standardized

and structurally dense than the unique, personal tone of a CEO letter. Moreover, a lot of current techniques, such as the most recent LLM-based holistic scoring, generate "black box" answers or single composite scores that mix up many risk categories. They are therefore inappropriate for assessing the particular assumptions of this study, which need separating rhetorical accountability from disclosure quality.

In order to close this gap, the GWRI was created as a sentence-level, multidimensional tool especially calibrated for the particular genre of the CEO letter. In contrast to the ESG disclosure-performance gap, which determines whether a discrepancy occurs, the GWRI describes the linguistic construction of that gap. The index offers the detailed, variable-level information required to connect particular communication tactics to both external ESG ratings and ensuing market performance by breaking down greenwashing risk into three theoretically supported dimensions: Substantive Disclosure Quality, Commitment Accountability, and Rhetorical Positioning. The three subdimensions correspond directly to the theoretical constructs reviewed in Chapter 2: Substantive Disclosure Quality (grounded in signaling theory and impression management), Commitment Accountability (grounded in decoupling theory and the Gricean Cooperative Principle), and Rhetorical Positioning (grounded in Systemic Functional Linguistics and the Appraisal Framework). While the GWRI concept consists of seven-variable index (C1-C7), organized in the subdimensions, the final operational version used for empirical research is a reduced-form implementation that only includes the rule-based variables. For technical specifications of the tool, see Appendix D. The GWRI was implemented in Python 3.13 through a modular pipeline: environmental sentence identification (Module 3), variable scoring (Module 4), winsorization and standardization (Module 5), and composite aggregation (Module 6). The full code is provided in Appendix C.

The identification and classification of environmental sentences were supported by a large language model-assisted procedure. This approach was used exclusively for text preprocessing and segmentation, while all final variable constructions (C1, C3, C4, and C7) are based on deterministic, rule-based criteria to ensure full reproducibility and consistency across observations. The rhetorical dimensions grounded in SFL and the Appraisal Framework, covering claim quality (C2), attitudinal intensity (C5), and engagement patterns (C6), are theoretically motivated in Chapter 2 but excluded from the operational implementation for reasons of reproducibility, as explained in Section 3.5.1.

3.5.1 Greenwashing Warning Risk Variables

The Claim Specificity Index (C1) is designed to measure how verifiable and informational the environmental claims are at the sentence level. It lies primarily in the signaling theory, which suggests that a signal is only credible if it is costly to fake (Connelly et al., 2011, p. 39). A company that makes ambiguous, unquantifiable sustainability claims in CEO letters does not suffer reputational harm because there is nothing concrete enough to be held responsible for. In contrast, a company that reports Scope 1 emissions reductions against a defined baseline with a year objective establishes a verifiable promise that can be audited by the market and ESG rating organizations. C1 effectively conveys the distinction between these two categories of claims. In this study, each environmental statement is rate on a scale of 0 to 4, where 4 necessitates a full set of requirements, such as measurable facts, accepted standards, a clearly defined scope and a deadline, while a 0 indicates a pure assertion with no supporting evidence. We compute the average of all environmental sentences to get the overall document score. However, we flip the C1 value before combining it into the final composite because a high specificity score implies a decreased probability of greenwashing. This modification

guarantees that a larger numerical value consistently indicates a higher risk level across all study variables.

Table 2: Claim Specificity Index

Score	Evidential Level	Example
4	Evidence + scope/baseline + year target	Reduced scope 1 CO_2 by 23% against at 2018 baseline, targeting net zero by 2030
3	Evidence + scope/baseline or evidence + year target	Reduced scope 2 emissions by 40% compared to 2019 levels
2	Quantified or named standard only	Aligned with SBTi; carbon intensity declined by 18%
1	Vague commitment without specific evidence or target	We are working at reducing our carbon footprint
0	Pure assertion	We are deeply committed to a sustainable future

Moreover, Action Temporality Ratio (C3) captures the balance between forward-looking commitments and evidence of completed or ongoing activities. Looking from a finance and agency theory perspective, a CEO letter dominated by future commitments with little evidence of past delivery is a low-cost legitimacy claim, where management signals environmental ambition without facing any immediate accountability. This is the linguistic manifestation of decoupling, where firms adopt aspirational narratives to satisfy stakeholder expectations while leaving underlying operations unchanged (Meyer & Rowan, 1977, p. 356-359). Therefore, when future language predominates, investors and ESG rating agencies that depend on CEO letters as an additional signal are subject to a systematic upward bias in perceived sustainability commitment. To measure this, we classify each environmental sentence based on its tense and intent. Sentences that include "will," "plan to," or "intend to" are marked as future-oriented, whereas sentences that use static numerical disclosures or past-tense formulations are marked as finished. The ratio of these prospective commitments to actual accomplishments is the final document score. A larger ratio in this model is a warning indicator for possible greenwashing danger because it shows a substantial temporal imbalance. From a linguistic perspective, a CEO letter dominated by future commitments with minimal evidence of current action also constitutes a systematic violation of Grice's Quantity maxim, the communicative expectation that a speaker provides a sufficiently complete account, making paltering through selective temporal emphasis a direct manifestation of the greenwashing risk C3 is designed to capture.

If the previous metrics identify that a commitment is being made, the Commitment Horizon Distribution (C4) assesses how much near-term discipline that promise imposes management. The finance relevance here is straightforward, a commitment to achieve net zero by 2050 made in a 2023 letter imposes no near-term market discipline on management, while a commitment to reduce Scope 1 emissions by 20% by 2025 creates genuine accountability risk within the investment horizon of most institutional investors. Firms that systematically make long-horizon or vague commitments can maintain the appearance of sustainability leadership while deferring any real obligation indefinitely, which is a structural form of what the greenwashing literature identifies as paltering (Janik & Ryszko, 2025, p. 3). Only future-classified sentences from C3 are used, each classified by its offset from the letter year as short-term at two years or fewer, medium-term at three to nine years, long-term at ten or more years, or vague where no year is identified. The C4 score is the share of future commitments falling in the long-term or vague categories. Critically, classification is always relative to the letter year rather than

absolute, so a 2030 target is short-term in a 2028 letter but long-term in a 2022 letter, ensuring comparability across the sample period. In Gricean terms, long-horizon or vague commitments violate the Manner maxim by obscuring the clarity and verifiability of the stated promise, which is precisely the paltering mechanism identified by Lyon and Montgomery (2015, p. 224) as a core linguistic feature of greenwashing.

Furthermore, Structural Emphasis Score (C7) measures where, on average, environmental sentences appear within the CEO letter. The finance relevance of this variable lies in investor attention research, where content appears in a document to shape how much attention it receives, independently of what the content says (Merkl-Davies & Brennan, 2007, p. 119-120). A firm that places its environmental commitments prominently at the front of the CEO letter is signaling that sustainability is a genuine strategic priority. A firm that buries environmental content toward the back is more likely, as a compliance exercise or reputational hedge rather than as a substantive communication to investors. C7 is the arithmetic mean of normalized positional values of all environmental sentences, where 0.0 indicates front-loaded content and 1.0 indicates back-loaded content. It is treated as a supplementary structural indicator in the composite and assigned a lower weight than the content-based variables, given that position alone is a weaker signal than claim quality or temporal accountability.

Finally, it is important to note that the full GWRI model also includes three additional variables which are C2, C5, and C6 (See Appendix D). However, they were excluded from the operational version due to prioritizing reproducibility. These variables primarily target the rhetorical dimensions of greenwashing such as whether the language used was inflated, promotional, or evasive. These variables may not guarantee fully consistent scores across different computational environments due to their reliance on model-based interpretation. For this study, where scores feed directly into a regression, it is important to ensure stability and consistency. The four rule-based variables always produce identical scores for identical inputs, making every data point fully auditable. This is the most significant limitation of the empirical implementation and is discussed further in Section 3.9.

3.5.2 Composite Index Construction

Once the four active variables were computed, the final step involved combining them into a single GWRI composite score through a standardized aggregation process. Each variable was first winsorized at the 1st and 99th percentiles, which prevents the final index from being distorted by extreme outliers. After that, these values were standardized using the z-score to put them on common scale, and C1 was flipped to ensure that a larger number always indicates a higher risk of greenwashing.

The main method to calculate the composite is Version A, a weighted structure based on established the established theory. This method allocates value based on the empirical evidence found in current literature, as opposed to allowing the data to determine the weights. In particular, 40% of the total weight is given to both Substantive Disclosure Quality and Commitment Accountability, which reflects their solid signaling and decoupling theoretical underpinnings. Rhetorical positioning receives the remaining twenty percent. In actuality, this means that C1 bears the entire 40% of the accountability weight for disclosure quality; C3 and C4 divide the 20% apiece, and C7 bears the remaining 20% for positioning. Importantly, these weights were established prior to any data analysis, guaranteeing that the model continues to be a theoretical reasoning test rather than a "data fitting" exercise (OECD & Joint Research Centre, 2008).

Furthermore, to ensure that these results are not only a result of the weights selected, but the study also incorporates a robustness check called Version B. Version B gives each of the four factors a straightforward 25% equal weight. Any notable difference between the two versions would indicate that the results are susceptible to our presumptions, necessitating a closer examination of the particular subdimension influencing the result. Lastly, although Principal Component Analysis (PCA) is a popular weighting option, it was purposefully ignored in this instance. PCA-derived weights vary based on the data utilized because they are designed to explain variance within a particular sample. This would make it challenging to compare our results with those of other studies. Additionally, PCA would be statistically underpowered and unduly sensitive to small sample fluctuations with a sample of 113 enterprises and only four variables. For this reason, theory-based weights continue to be a more dependable and understandable option.

3.6 Dependent Variables

3.6.1 ESG Ratings

The analysis derives firms´ sustainability performance from the LSEG Refinitiv ESG score as it is widely used and has over 180 firm level indicators across environmental, social and governance sections. The analysis only uses Refinitiv as ESG data provider for two main reasons. Firstly, Refinitiv offers more comprehensive coverage of Nordic publicly listed firms compared to the other alternatives like MSCI or Sustainalytics. Furthermore, only a single rating provider maintains consistency of the measurement. Nevertheless, the analysis is exposed to the provider´s methodological choices. Given that consensus framework for aggregating disparate metrics do not exist (Berg et al., 2022), any choice of the provider will reflect the broader industry inconsistencies. Therefore, combining scales across different providers would cause additional error without a defensible weighting scheme. Lastly, the ESG score is used as a dependent variable in regression H1 model and as a control variable in the H2 model.

3.6.2 Stock Market Performance

The dependent variable for H2 is the annual stock return, which is calculated as the percentage change in a company's closing price over a twelve-month period. Specifically, the annual return for year t is computed as:

$$\frac{(P_t - P_{t-1})}{P_{t-1}}$$

Where, P_t is the closing price of the end of year *t*. For this study, we have used raw returns rather than risk adjusted or benchmarked adjusted return for two main reasons. First, the study does not aim to test whether greenwashing risk generates alpha in the asset-pricing sense, but rather to assess whether a high linguistic risk signal in the CEO letter predicts weaker absolute equity performance in the subsequent year. Second, constructing risk-adjusted returns would require specifying a factor model (such as the Fama-French three-factor or five-factor model) and matching each Nordic firm to a local market benchmark, which would introduce additional methodological choices beyond the scope of this thesis. The use of raw returns is consistent with related studies that similarly examine the association between ESG-related disclosure characteristics and forward stock performance (Walker & Wan, 2012, pp. 227-240; Gorovaia & Makrominas, 2025, pp. 438-441).

To mitigate simultaneity bias and allow sufficient time for the market to assess whether CEO sustainability claims are substantiated, a one-year lag is introduced between the publication of the CEO letter and the measurement of stock performance. A twelve-month horizon is chosen because greenwashing is typically exposed through news coverage, analyst scrutiny, or failure to meet stated ESG targets within an annual reporting cycle, which allows the market to reassess the credibility of the prior year's sustainability narrative. It should be noted that this one-year horizon represents a conservative test of the greenwashing-return relationship: if the market reaction to narrative-level greenwashing operates through a slower channel, such as the gradual erosion of B2B business relationships or ESG-driven capital reallocation, effects may only become detectable over a two- to three-year horizon. The return alignment in the dataset is as follows:

- *2022 CEO letters are matched with 2023 stock returns*
- *2023 CEO letters are matched with 2024 stock returns*
- *2024 CEO letters are matched with 2025 stock returns*

3.7 Control Variables and Estimation Controls

This study does not control for firm-level financial characteristics such as size, leverage, profitability, or book-to-market ratio. This was a practical constraint given the data and timeline available for this research. Lack of these control variables means that some of the variation we attribute to greenwashing risk could reflect underlying firm characteristics in principle. We acknowledge this openly as a limitation and return to it in Section 3.9. That being said, the inclusion of sector and year fixed effects does absorb a meaningful share of the systematic differences across industries and time periods. The main regression H1 includes sector fixed effects and year fixed effects with clustered standard errors at the firm level. Industry fixed effects are implemented for stable differences between industries such as regulation levels, environmental intensiveness, and reporting approaches. On the other hand, the year fixed effect absorbs broad shocks that effect every firm in a given year as macroeconomic conditions or change in ESG rating calculations. Observations from the same firm across multiple years are modeled to be dependent by clustering standard errors at the firm level. Moreover, the analysis re-estimates the main specifications by winsorized greenwashing scores in order to reduce the influence of the extreme values and uses lagged one year greenwashing score to address potential concerns about reverse causality for robustness checks. We acknowledge that lack of firm level financial control can cause omitted variable bias when the results are interpreted. Additionally, the short length of 3 years also limited our ability to include firm fixed effects.

3.8 Empirical Model

3.8.1 Model Specification

The empirical model tests two hypotheses. The main hypothesis H1 is tested with the following model:

$$ESG_{i,t} = \alpha + \beta \cdot 1GWRI_{i,t} + \mu_{\text{sec}\,tor} + \tau_{year} + \varepsilon_{i,t}$$

where $ESG_{i,t}$ represent the LSEG Refinitiv ESG score for the firm *i* in the year *t*; $GWRI_{i,t}$ shows the results of Greenwashing Warning Risk derived from the CEO letter of frim *i* in year *t*; μ_{sector}

represents a vector of sector fixed effects; τ_{year} is a vector of year fixed effects and the error term is $\varepsilon_{i,t}$. Our primary coefficient of interest β1 is predicted to be negative according to H1.

The second hypothesis H2 has the following model:

$$Ret_{i,t+1} = \alpha + \beta_1 \cdot GWRI_{i,t} + \beta_2 \cdot ESG_{i,t} + \mu_{\text{sec}\,tor} + \tau_{year} + \varepsilon_{i,t}$$

Where $Ret_{i,t+1}$ represents the one year ahead stock return for the firm *i*. By the inclusion of ESG score as a control, β_1 effectively captures the narrative of greenwashing which captures the CEO rhetoric which deviates from the actual ESG performance. This specification aligns with H2′s rhetoric claim of concerning the gap between "the talk" (language of CEO letter) and "walk" (measured ESG performance). Under this hypothesis, we expect the coefficient of interest β_1 is predicted to be negative under H2.

3.8.2 Estimation Strategy

We tested both models by ordinary least squares since residuals of the same firm are likely correlated as each firm appears three times in the panel. Therefore, we clustered the standard errors at the firm level in all robustness specifications. Such an adjustment facilitates more conservative inference and aligns with established econometric practices for panel data (Petersen, 2009, pp. 435-436). As a result, there are two GWRI estimates. GWRI-A is implemented in the main specification, and it applies the theoretically motivated weighting of 40% to Substantive Disclosure Quality (C1), 40% to Commitment Accountability (C3 and C4), and 20% to Rhetorical Positioning (C7). GWRI-B is an equal-weighted variant for estimating in parallel as a robustness check on the weighting choice. All empirical analyses were conducted using *Stata* (version 19.5), including fixed effects estimations and clustered standard errors.

Looking back at this chapter, the methodological choices did not come together by coincidence. One decision shaped what was possible for the next decision. Committing to a positivist stance early on meant the study had to work with observable, quantifiable data. Hence, it closed off subjective or interpretive approaches before they were even seriously considered. From this point of view, a deductive approach was not simply preferable but necessary given that the research already had theoretically grounded hypotheses which needed empirical testing rather than theory to be built from scratch. Once the study was committed to deductive, quantitative design with firm level data spanning three years, panel methods were not something we chose so much as something the data structure required. Within that setting, OLS with clustered standard errors and fixed effects was the appropriate choice for handling firm correlation and sector level differences. The decision to develop GWRI, rather than adopting an existing tool followed from the same reasoning. When we specified the hypotheses, we found that no available instrument could capture the rhetorical constructs that were required, and building the index was therefore not a preference but a response to that gap. Taken together, the design of this study reflects a chain of reasoning where each step is constrained and informed by the previous one.

3.8.3 Robustness Checks

The robustness assessment is applied through four complementary specifications. Firstly, we aggregated 47 industry fixed effects (Material & Chemicals, Energy, Technology, Industrials, Healthcare, Transportation, Consumer, and Services/Other) to preserve statistical power and avoid overfitting, given the presence of sparse industry clusters within our limited sample.

Additionally, the forward return and GWRI are winsorized at the 1st and 99th percentiles to ensure that our estimates are not driven by spurious outliers. Third, we introduce firm fixed effects to capture all time invariant firm characteristics as a replacement of the broader sector effects. Lastly, the analysis lags the GWRI variable by one period to examine whether the link ESG performance and greenwashing risk holds under temporary separation. The approach creates a safeguard against reverse causality such as the possibility that poor ESG ratings lead firms to adopt more defensive or exaggerated language that can be flagged as green washing in CEO letters.

3.9 Validity, Reliability & Limitations

This study has several limitations that need to be acknowledged. First, the GWRI measures the linguistic features in CEO letters instead of confirmed greenwashing behavior. High greenwashing signals that the CEO letter contains language patterns which are theoretically associated with greenwashing. Therefore, a company with high GWRI does not necessarily engage in actual greenwashing. Conversely, there can be cases where a firm engaged in greenwashing may write CEO letters that avoid the specific linguistic markers our index captures. For this reason, the index should be interpreted as a measure of linguistic risk rather than a direct measure of deceptive corporate behavior.

Secondly, we implemented four of the seven variables originally theorized in our framework (C1, C3, C4, C7) for guaranteed identical scores across runs. The three remaining variables target the rhetorical and appraisal dimensions of greenwashing (C2 on claim quality, C5 on attitude, and C6 on engagement) and require LLM based classification which may not provide same results across different runs. Accordingly, we prioritized reproducibility over implementation coverage. As a result, our index may underweight the purely rhetorical dimensions of greenwashing due to absence of LLM.

Third, our research only relies on a single ESG rating provider (LSEG Refinitiv). Previous studies illustrate divergence across ESG rating providers conditioned by differences in their scope and aggregation rules (Berg et al., 2022). Consequently, our results may differ under alternative ESG rating sources. On this basis, our conclusions regarding H1 should be interpreted as conditional on the Refinitiv rating methodology. Fourth, our selection is based on English language CEO letters, and the analysis may introduce selection bias against firms that publish CEO letters only in their native languages. Hence, the results overrepresent larger and more internationalized firms. Additionally, our sample is composed of 113 firms observed over three years. The sample and duration are sufficient for identifying cross-sectional patterns yet limited for detecting dynamics within the firm. Correspondingly, the firm's fixed effect specifications have limited statistical power. Fifth, we excluded firm level financial controls such as firm size, leverage, profitability, book-to-market ratio, prior stock return due to time and data access constraints during the research timeline. The analysis partially addresses this issue through industry and year fixed effects in the main specifications, and by incorporating firm fixed effects in robustness specifications. However, firm size and capital structure within industry variation remain uncontrolled in our model. Therefore, this limitation should be considered when results are interpreted, and future research can extend our analysis with broader data.

Beyond these points, it should be noted that the composite weighting of the main index GWRI-A reflects theoretical reasoning (40/40/20) rather than data driven optimization. On the other

hand, equal weighted GWRI-B specification provides a sensitivity check on this choice, yet the theoretical weights themselves are not empirically validated. Lastly, the use of a one year forward return horizon may be insufficient to capture market responses to greenwashing signals to longer-term. In cases where market participants price risk only upon discrete public exposure (regulatory actions, journalist investigations, NGO reports) the results do not coincide with our annual observation window. Event study designs and longer horizon studies would complement our analysis.

3.10 Ethical and Societal Considerations

It is important that the data used for research is not manipulated or fabricated in order to achieve desired outcomes. This study relies exclusively on publicly available data, including CEO letters and ESG ratings, and therefore does not involve confidential or personal information. As such, no direct ethical risks related to data privacy or human subjects arise. All data has been obtained in a lawful and transparent manner. In addition, individual firms are not explicitly identified or discussed in relation to extreme Greenwashing Warning Risk Index (GWRI) values in the analysis. This approach is adopted to ensure that the study remains focused on aggregate patterns rather than firm-level reputational judgements.

Artificial intelligence (AI), in the form of a large language model (Claude Code) was used in a limited and supportive capacity at two specific stages of the research process. First of all, plan mode of Claude Code is used as a helper when coding the tool. Additionally, it assisted in identifying and segmenting environmental sentences within CEO letters, classifying each sentence as environmental or non-environmental based on a predefined keyword framework. Secondly, it provided preliminary classification support for sentence temporality, helping distinguish between future-oriented and completed action sentences as an input to the C3 variable construction. In both cases, the LLM outputs served as preprocessing inputs only and were not used directly as final variables. Importantly, all final variables used in the empirical analysis are constructed using deterministic, rule-based procedures, ensuring full transparency, reproducibility, and researcher control over the analytical framework. AI tools were not used in the writing of this thesis, the collection or interpretation of data, or the statistical analysis of results. Academic integrity and student independence were maintained throughout the research process, and the complete GWRI code is provided in Appendix C for full transparency and reproducibility.

We recognize the ethical responsibility in financial research to ensure accurate, transparent, and honest presentation of data and results. Any limitations of the study, including those related to measurement uncertainty, data quality, and methodological constraints, are openly acknowledged and discussed in the relevant sections of the thesis. The study is conducted with the aim of contributing to academic knowledge with integrity, transparency, and methodological rigor.

4. Empirical Findings

Chapter 4 will present and interpret the empirical findings of the study including the descriptive and correlation analysis. In addition, this chapter presents a full set of regression results for Hypothesis 1 and the corresponding results for Hypothesis 2. Finally, it evaluates all four robustness checks embedded in the empirical strategy. This chapter serves as a foundation for discussion in chapter 5.

4.1 Descriptive Statistics

As previously mentioned in chapter 3, the final estimation sample consists of 113 B2B firms from the Nordic countries, covering fiscal years 2022, 2023, and 2024 All these 113 firms contribute three annual observations, giving an unbalanced panel of 339 observations used in the contemporaneous specifications. Since the first year of each firm panel (2022) is used to build the one-year-lagged variable, the effective sample for the lagged GWRI specifications in column (5) of Tables 5 and 6. is reduced to 226 firm-year observations. The sample-construction process as outlined in Section 3.4.3, is consistent with the panel's solid balance within the three-year window. The table below (Table 3) shows the descriptive statistics for the five variables used in the regression analysis.

Table 3: Descriptive statistics of the five variables

Variable	N	Mean	SD	P25	Median	P75	Min	Max
GWRI-A	339	−0.009	0.504	−0.331	0.001	0.318	−1.906	1.581
GWRI-B	339	−0.001	0.530	−0.359	0.009	0.368	−2.118	1.538
ESG score	339	60.905	16.313	48.988	63.590	74.273	18.514	91.493
Return (t)	226	0.079	0.297	−0.136	0.053	0.265	−0.624	1.750
Return (t+1)	339	0.099	0.301	−0.103	0.064	0.275	−0.624	1.750

Notes: GWRI-A is the theory-weighted Greenwashing Warning Risk Index (40 % Substantive Disclosure Quality, 40 % Commitment Accountability, 20 % Rhetorical Positioning). GWRI-B is the equal-weighted variant. Both indices are standardised z-scores before composite construction. The ESG score is the LSEG Refinitiv combined ESG score on a 0-100 scale. Return (t) is the contemporaneous raw annual stock return; Return (t+1) is the one-year-ahead raw annual stock return used as the dependent variable in the H2 regressions.

GWRI-A, the theory motivated model, and GWRI-B, the equal-weighted variant, demonstrate a high level of distributional similarity, mainly due to the fact that they are constructed from the same underlying sub-indicators. Notably, both versions lie very close to zero, with a mean of -0.009 and -0.001 respectively. These results do not provide an absolute measure of firms' greenwashing, instead, this near-zero centering supports the standardization and winsorization procedures by demonstrating that the index accurately represents a firm's relative position within the sample. Furthermore, the two index's dispersion is roughly almost the same, although GWRI-B has a slightly higher standard derivation (0.530 vs. 0.504). This small difference indicates that sub-indicators with lower weights in the theoretical specification are somewhat more sensitive to the equal-weighted model. As for the distributional shape, the median (0.001) for GWRI-A is nearly identical to the mean. This suggests that the data is essentially symmetric.

The inter-quartile range does span from -0.331 to 0.318 and represents a total spread of roughly 0.65 standardized units. This spread is considered wide enough to suggest that a “typical” firm-year observation differs significantly from its counterparts on the opposite side of the median. Moreover, the presence of relatively wide tails with a minimum of -1.906 and a maximum of 1.581 highlights the existence of extreme cases. represent firm-years in which the CEO letter scored, respectively, exceptionally well or exceptionally poorly across the four sub-dimensions simultaneously. For example, firms at the upper tails of the GWRI-A distribution represent “worst-in-class”. Specifically, these high-scoring observations characterize letters that are vague in their environmental claims (C1), disproportionately future-oriented (C3), horizon-deferred in their commitments (C4), and finally, structurally obscured, with environmental content placed toward the end of the document (C7).

The ESG score variable has a mean and standard deviation of 60.905 and 16.313 respectively. The median is 2.7 points above the mean, 63.590 to be specific, which indicates a modest negative skewness. This suggests that while the majority of large Nordic listed companies cluster in the mid-to-upper range of Refinitiv’s scoring system, a small group of lower-performing firms pulls the average down. The inter-quartile range of the ESG score does span from 48.988 to 74.273 with a spread of roughly 25.3 points. This difference suggests that the middle 50% of the sample covers more than 1.5 standard deviations. Furthermore, the full range, from 18.514 to 91.493, reveals that even within the high-sustainability Nordic region, there is significant variation in how external agencies assess performance. This heterogeneity is essential for the study's identification strategy, as meaningful cross-sectional variation is required to ensure that the regression coefficients remain precise regardless of the quality of the greenwashing measure.

In terms of stock returns, it follows typical distributional patterns found in equity data. The one-year-ahead return (ret_fwd) has an inter-quartile range between −10.3% and 27.5%, with a mean of 9.9% and a standard deviation of 30.1%, and a median of 6.4%. Since the maximum return of 175.9% is much further from the median than the minimum of -62.4%, the distribution shows a positive skewness, which reflects the standard asymmetry of equity returns, where losses are capped at 100% while gains are theoretically infinite. Consequently, this skewness is particularly relevant for the Hypothesis 2 analysis, since the OLS regression can be sensitive to right-tail outliers, a factor that motivated the winsorization check in Table 6 (column 4). In addition, the 9.9% mean return reflects a sample period (2022-2025) marked by the 2022 market downturn driven by macroeconomic headwinds including energy crisis and monetary tightening, the 2023 partial recovery, and continued growth into 2024. This macroeconomic volatility is absorbed by year fixed effects, which explain why year controls affect the intercept but leave the GWRI coefficient largely unchanged. Finally, the contemporaneous return (ret) has a slightly lower mean of 7.9% and is limited to 226 observations, as 2025 data is not yet fully available for the most recent letters.

4.2. Correlation Analysis

Table 4 shows the Pearson correlation matrix for all five variables. The matrix provides an overview of bivariate correlations before controlling the sectoral and temporal heterogeneity. Moreover, several patterns are useful for the subsequent analysis of regression.

Table 4:Pairwise correlations

	GWRI-A	GWRI-B	ESG score	Return (t)	Return (t+1)
GWRI-A	1.000				
GWRI-B	0.961***	1.000			
ESG score	−0.152**	−0.135*	1.000		
Return (t)	−0.038	−0.047	−0.036	1.000	
Return (t+1)	0.048	0.051	0.006	0.149*	1.000

*Note: The Pearson correlation coefficients *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels respectively. The number of observations is N = 339 for correlations not involving Return (t) and N = 226 for correlations involving Return (t).*

The results show a nearly perfect correlation between GWRI-A and GWRI-B ($\rho = 0.961$, $p < 0.001$), indicating strong measurement consistency. The two indexes differ in their aggregation weight, but the underlying components are identical. Therefore, the high correlation demonstrates that firm observations by greenwashing risk are largely insensitive to changes in weighting schemes. The finding is important because composite indices in social sciences often display sensitivity to choices in aggregation. Furthermore, the results provide evidence that the underlying sub-indicators (C1, C3, C4, C7) are positively correlated and have a tendency to move in the same way across different observations. Therefore, alternative weighting structures can produce very similar composite scores. This internal consistency strengthens the empirical credibility of the GWRI.

GWRI-A and the ESG score correlation ($\rho = -0.152$, $p < 0.05$) confirms the bivariate direction of the relationship that was examined in H1. The results indicate that higher levels of greenwashing risk in CEO letters are associated with lower Refinitiv ESG scores on average. There is a modest association with a magnitude of -0.152 and the result is consistent with the expectation that the CEO letters as a communication with stakeholders represents only a single component of the broader information source that is used by rating agencies. However, the 5% statistical significance shows that the relationship reflects a systematic pattern instead of a result driven by a small number of observations. The corresponding correlation for GWRI-B ($\rho = -0.135$, $p < 0.10$) is slightly weaker but directionally consistent. Together, these findings suggest that the observed relationship is robust to alternative weighting methods.

On the other hand, the results do not show any significant correlations between the GWRI and stock returns as the numbers are close to zero. For returns made one year in advance, the correlations are $\rho = 0.048$ for GWRI-A and $\rho = 0.051$ for GWRI-B, both of which are statistically indistinguishable from zero. Since neither estimate is significant, no directional interpretation can be drawn from them. Although the correlations do not control for year or industry effects, the absence of any detectable bivariate relationship provides no initial support for H2, which will be further assessed in the multivariate regression analysis.

Another noticeable feature is the extremely low correlation between ESG scores and one year ahead returns ($\rho = 0.006$, $p > 0.10$). It is expected to observe a positive association with future returns if the ESG scores fully capture relevant sustainability performance. Nevertheless, broader empirical research shows the absence of such consistency connection, and our result is in line with these. For instance, Berg et al. (2022, p.1318) documented that the divergence between different ESG rating providers reduces the content of information of ESG scores, and Friede et al. (2015, p. 211) highlighted that financial performance and ESG scores are context

dependent and mostly insignificant in the short-term horizons. Hence, the null ESG score and financial return correlation in our sample is not unexpected, yet it provides a useful benchmark for interpreting the H2 results. Finally, it is worth noting that contemporaneous returns and one year ahead returns are significantly but modestly correlated ($\rho = 0.149$, $p < 0.05$). This pattern provides additional context for the return dynamics observed in the sample.

4.3 Hypothesis 1: Greenwashing Risk vs ESG ratings

A full set of the regression estimates can be shown in table 5 below, with the Refinitiv combined ESG score serving as the dependent variable across all five specifications. The analysis begins in Column (1) with a baseline pooled OLS regression, where GWRI-A serves as the sole explanatory variable. Column (2) builds upon this by introducing year fixed effects. This feature is essential for absorbing common time-series shocks, including more general changes in the global ESG regulatory environment or recurring adjustments to Refinitiv's scoring methodology. Moreover, Column (3) represents the study's preferred specification. In addition to year fixed effects, this model incorporates sector fixed effects and utilizes standard errors clustered at the firm level. This approach accounts for industry-specific characteristics and addresses potential within-firm serial correlation in the residuals over the three-year panel. Finally, Columns (4) and (5) serve as robustness checks to ensure the stability of the primary findings. Specifically, Column (4) employs a version of the GWRI-A winsorized at the 1st and 99th percentiles to mitigate the influence of extreme values. Column (5) shifts the temporal focus by replacing the contemporaneous GWRI-A with its one-year lag, testing whether the observed relationships persist when considering delayed market or agency reactions.

Table 5: H1 - Greenwashing Warning Risk Index and ESG Ratings

Dependent variable: ESG score	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**
	Pooled OLS	+ Year FE	+ Sector FE	Winsorised	Lagged GWRI
GWRI-A	−4.918***	−4.923***	−4.715**		
	(1.759)	(1.760)	(1.961)		
GWRI-A (winsorised)				−4.765**	
				(2.051)	
GWRI-A (t−1)					−5.338**
					(2.362)
Year FE	No	Yes	Yes	Yes	Yes
Sector FE	No	No	Yes	Yes	Yes
Clustered SE	No	No	Yes	Yes	Yes
Observations	339	339	339	339	226
R^2	0.023	0.023	0.112	0.112	0.118

*Notes: The dependent variable is the LSEG Refinitiv combined ESG score (0-100). GWRI-A is the theory-weighted composite index. Winsorised GWRI-A is winsorised at the 1st and 99th percentiles. GWRI-A (t−1) is the one-year lag of the composite. Standard errors are in parentheses; in columns (3)-(5) they are clustered at the firm level. *, **, and *** denote statistical significance at the 10 %, 5 %, and 1 % level, respectively.*

The preliminary analysis of H1 begins with the most parsimonious model in Column (1). Here, the GWRI-A coefficient is estimated at −4.918 with a standard error of 1.759, significant at the 1% level (p = 0.006). The meaning is straightforward: regardless of any other observable trait, a one-unit rise in the composite greenwashing risk index is linked to a 4.92-point decrease in the ESG score in the same firm-year. The GWRI alone accounts for around 2.3% of the sample's cross-sectional and temporal variation in ESG scores, according to the R2 of 0.023. It proves that the language characteristics of a single, somewhat short element of the annual report, in this case the CEO letter, contains sufficient "signal" to forecast the intricate, multifaceted evaluation of a specialist third-party rating agency.

Building on the initial model, Column (2) introduces year fixed effects that serve to isolate the association from aggregate temporal shocks, with an unchanged coefficient at -4.923, and no material increase in R2. This constancy across columns (1) and (2) is a useful internal validity check since it implies that the observed link is not a result of more general market trends, like a year-over-year shift toward more "aspirational" discourse across all firms or recurring changes to Refinitiv's internal scoring algorithms.

The analysis further refines the identification strategy in Column (3) by incorporating sector fixed effects and clustering standard errors at the firm level. This is the study’s preferred specification, as it addresses the systematic differences in environmental profiles and reporting cultures inherent to different industries. Here, the R2 increases with nearly 9 percentage points, from 0.023 to 0.112. This significant increase confirms that the industry is a dominant predictor of ESG scores, and reflects ingrained variations in reporting cultures, regulatory exposure, and environmental footprints among industries. Crucially, however, the GWRI-A coefficient remains robust. Despite the inclusion of these industry controls and the application of firm-level clustered standard errors, it remains statistically significant at the 5% level (p=0.018), where the coefficient only modestly attenuates to −4.715 from -4.923. This persistence is especially relevant for the thesis, since it shows that GWRI captures within-sector variation. Simply put, when comparing two firms in the same industry during the same year, and therefore holding the external macro-environment constant, their ESG ratings still differ in a direction predictable by their CEO letters. These supports signaling and impression management theories, suggesting that greenwashing risk is a deliberate, firm-specific rhetorical choice rather than a mere reflection of industry-level reporting norms.

To move beyond statistical significance, the economic magnitude of the findings requires careful calibration. Using the suggested estimate from Column (3), a one-standard-deviation increase in the GWRI-A (0.504 units) is expected to lead to a drop of approximately 2.38 ESG points. Given that the sample's ESG score standard deviation is 16.313, this effect size translates to roughly 14.6% of a standard deviation, or one-seventh of the average spread between firms. Additionally, according to the GWRI's inter-quartile range of 0.649 units. The cumulative effect could be significant even though a 3.1-point difference might not instantly move a company across a significant rating barrier (such going from a B to an A). The implicit penalty is estimated to approximately 9.3 points, if a firm’s communication approach consistently puts it in the high-risk quartile over a three-year period. If a shift arises, this might result in a tier downgrade under the Refinitiv framework, which could have an impact on the term of credit facilities linked to sustainability or lead to exclusion from ESG-screened investment mandates.

To verify that the results are not sensitive to outliers, Column (4) employs a winsorized version of the index. This guarantees that the results are not being disproportionately driven by extreme observations in the tails of the distribution. The resulting coefficient (−4.765) is almost exactly the same as the full-sample estimate, indicating that the connection is widespread within the sample's normal range of firms. Lastly, Column (5) tackles the main issue of reverse causality, which is the potential for companies with low ESG scores to react by using more ambiguous or defensive language. We separate the relationship between previous rhetoric and future ratings by using a one-year lag of the GWRI-A. In comparison to the contemporaneous finding, the projected coefficient of −5.338 is both significant and greater in size. This implies that rating agencies may take some time to fully process the "greenwashing signal" in a CEO letter, or that the lag structure is more in line with the agencies' own information cycles. Regardless of the precise mechanism, the evidence from all five specifications supports H1 coherently and strongly, demonstrating a direct connection between rhetorical risk and external sustainability evaluations.

4.4 Hypothesis 2: Greenwashing Risk vs Subsequent Stock returns

Table 6 presents the regression results of H2. The one year ahead raw stock return is the dependent variable, and it is defined as the annual return in year t + 1 for a CEO letter that was published in year t. The Column (1) represents the pooled OLS regression of the forward return on the GWRI-A. The Column (2) adds the Refinitiv ESG score as a control, and the GWRI-A coefficient can be explained as capturing the additional predictive content of CEO language in relation to the firm's external ESG rating. The Column (3) represents year and sector fixed effects with firm level clustered standard errors. The last two columns (4 and 5) use the same robustness logic as in the previous table (table 5), using the winsorized GWRI and lagged GWRI.

Table 6: H2 - Greenwashing Warning Risk Index and subsequent stock returns

Dependent variable: Return (t+1)	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**
	Pooled OLS	+ ESG	+ FE	Winsorised	Lagged GWRI
GWRI-A	0.029	0.030	0.026		
	(0.033)	(0.033)	(0.033)		
ESG score		0.000	0.000	0.000	−0.000
		(0.001)	(0.001)	(0.001)	(0.001)
GWRI-A (winsorised)				0.018	
				(0.033)	
GWRI-A (t−1)					0.084
					(0.061)
Year FE	No	No	Yes	Yes	Yes
Sector FE	No	No	Yes	Yes	Yes
Clustered SE	No	No	Yes	Yes	Yes
Observations	339	339	339	339	226
R^2	0.002	0.003	0.031	0.031	0.048

Notes: The one year ahead raw annual stock return is the dependent variable. GWRI-A is the theory-weighted composite index. GWRI-A (winsorized) is winsorized at the 1st and 99th

percentiles. GWRI-A (t-1) is the one year lagged value of the index as used for adressing reverse causality concerns. The LSEG Refinitiv combined ESG score is included as a control in columns (2)-(5). The standard errors are in parenthesis and in the columns (3)-(5) they are clustered at the firm level.

H2 results display strong contrast to the findings for H1 since the coefficient on the GWRI is statistically indistinguishable from zero in every specification. The pooled OLS column (1) exhibits an estimate of 0.029 with a standard error of 0.033 ($p \approx 0.38$). Addition of the ESG score in the column (2) does not create any change at 0.030. The preferred specification with full fixed effects and clustered standard errors in column (3) produces 0.026 ($p \approx 0.43$), and the R^2 of 0.031 indicates that the full model, including sector and year dummies, explains only 3.1% of the variance in one-year-ahead returns. This limited explanatory power is itself informative: sector and year controls absorb almost no return variation either, confirming that in this sample the annual cross-section of equity returns is dominated by firm-specific shocks unrelated to sustainability measures. The column (4) that represents winsorized specification showcases even a smaller coefficient of 0.018. Meanwhile, the one-year lagged specification in column (5) gives the largest point estimate at 0.084 ($p \approx 0.17$). However, neither of the five estimates reaches a conventional level of statistical significance.

One feature of this result pattern deserves explicit discussion. The GWRI coefficient is statistically indistinguishable from zero across all five specifications. The 95% confidence interval for the true coefficient in the preferred specification spans approximately −0.040 to +0.091, encompassing zero in both directions. Because the estimates are not statistically significant, neither their sign nor their magnitude can be interpreted as meaningful evidence for or against any directional hypothesis. The relevant conclusion is simply that the data contain no reliable signal linking GWRI to one-year-ahead stock returns.

Second, the ESG score coefficient is rounded to 0.000 in every specification in which it is included, with standard errors of 0.001. The absence of any detectable return effect therefore extends beyond the narrative GWRI to encompass the fully processed, externally validated ESG rating itself. This pattern is consistent with Friede et al. (2015, p. 211), whose meta-analysis of over 2,000 studies indicates that the ESG-financial performance relationship is highly context-dependent and often reported as ambiguous or inconsistent across empirical studies, and with Berg et al. (2022, pp. 1316-1320), who show that provider divergence in ESG ratings injects substantial informational noise. The one-year return horizon used here may be insufficient to capture the greenwashing penalties theorised in Chapter 2: both the B2B relationship-damage channel identified by Walker and Wan (2012, pp. 227-240) and the ESG-driven capital reallocation channel described by Hartzmark and Sussman (2019, pp. 2791-2792) are inherently slow-moving mechanisms that operate over multi-year horizons. Hypothesis 2 is not supported by the evidence.

4.5 Robustness Check

We conducted four robustness exercises in the empirical strategy, and the results are reported in the main specifications on Tables 5 and 6. The robustness results are discussed jointly here in the context of their establishments and remaining limitations.

The first robustness dimension is the sector aggregation. Our sample data of granular Refinitiv industry classification consists of 47 distinct industry categories which many of them contain

only three firm and year observations. Therefore, retaining all 47 categories would produce a severely overfit model with a high possibility of producing clustered standard errors that are not converging in sparsely populated categories. Our analysis followed the convention recommended by Petersen (2009, pp. 435-436) by aggregation into eight broad sector groups: Materials & Chemicals, Energy, Technology, Industrials, Healthcare, Transportation, Consumer and Services/Other to avoid combined problem of overfitting and loss of degree of freedom. The GWRI-A coefficient under aggregated classification (-4.715 in Table 5, and 0.026 in Table 6) has the same order of magnitude and keeps the same statistical significance pattern as the unreported specifications under using all the 47 Refinitiv sectors. Therefore, the data shows that H1 regression result is not an artefact of coarse sector aggregation.

The second winsorisation in column (4) of both tables is applied for determining the possible effect of a small number of unusual CEO letters. For example, very short texts with high concentration of vague commitments or extreme returns driven by idiosyncratic firm level events. Both tables display results that winsorised estimated are close to its non-winsorized counterparts (-4.765 vs. -4.715 for H1; 0.018 vs. 0.026 for H2). The results indicate that the main findings are not generated by tails of the distribution that are outliers.

Moreover, column (5) represents the one-year lagged specification, and it addresses a reverse causality concern: a low ESG rating or a weak stock return might cause a change in the rhetorical style of the subsequent CEO letter. The lagged coefficient for H1 (−5.338) preserves the significant negative association found in the main specification, adding further confidence to that finding. For H2, the lagged coefficient of 0.084 remains statistically indistinguishable from zero, consistent with the absence of a detectable relationship across all other specifications. However, it should also be considered that the sample shrinks to 226 observations since the first-year data is absorbed into the lag. Crucially, the loss of observations is expected; the fact that H1 coefficient is significant at the 5% level in the smaller sample adds confidence to the results. The final robustness layer is provided by the high correlation between GWRI-A and GWRI-B ($\rho = 0.961$) which was documented in Section 4.2. Re-estimating both main specifications with GWRI-B in place of GWRI-A yields coefficients of −4.603 (SE = 1.954, $p < 0.05$) for H1 and 0.031 (SE = 0.034) for H2, preserving the significance pattern and approximate magnitude of the H1 finding, and confirming the absence of a statistically detectable effect for H2. This confirms that the results are driven by the underlying variation in the four sub-indicators rather than by the specific 40/40/20 weighting structure, and that the theoretical weighting choice does not materially influence the empirical conclusions.

In summary, the negative association between greenwashing risk in CEO letters and third-party ESG ratings survives across all specifications: with and without fixed effects, with clustered standard errors, with winsorization, with a one-year lag, and with an alternative equal-weighted index. The absence of a detectable effect on subsequent stock returns is equally robust across all five specifications, with the GWRI coefficient remaining statistically indistinguishable from zero throughout. These two headline findings provide the empirical foundation for the theoretical interpretation in Chapter 5.

5. Analysis and Discussion

This chapter develops a theoretically grounded interpretation of the empirical findings presented in Chapter 4, situating them within the broader academic literature on greenwashing, ESG information processing, and the market pricing of corporate sustainability. Furthermore, this chapter will discuss the overall contribution of the study within the relevant streams of theoretical and empirical literature.

5.1 Greenwashing Risk and ESG Ratings: A Multi-Theoretical Interpretation of H1

5.1.1 Signaling Theory and the Role of Costly Claims

The foundation of Signaling Theory that was formalized in the context of labour markets by Spence (1973, pp. 356-374), conveys that credible communication requires signals that are costly enough to prevent low quality agents from mimicking high-quality ones in the presence of information asymmetry. Connelly et al. (2011, p. 39) extends this framework to corporate contexts by explaining that a signal's credibility depends on three main properties: observability, cost, and clarity. In the setting of our research, sustainability claims in CEO letters are highly observable since the annual reports are a public document that reaches out to the full universe of capital market participants. In terms of clarity and cost, how they are constructed can influence heavily. A claim that is specific, quantified, and bounded in time, such as "We reduced Scope 1 emissions by 23% relative to our 2018 baseline, and we commit to net zero by 2030 through independently verified targets" is a costly claim to make since the firm faces reputational and legal risk if it fails to deliver it. Additionally, the claim is clear because the evidence required to verify the claim is well-defined. On the other hand, an aspirational assertion like "We are deeply committed to a sustainable future" is both cheap and vague as it carries no verifiable commitment. Therefore, the claim cannot impose no meaningful cost on the sender due to lack of verification regardless of actual performance.

The Claim Specifity Index (C1) forms the backbone of the disclosure quality dimension and precisely operationalizes this distinction. If the letter receives a low C1 score, it signals that the environmental sentences in the CEO letter predominantly rely on cheap, unverifiable claims, whereas a high C1 score signals that claims are supported by evidence and defined scopes. In fact, our finding that a higher GWRI predicts lower Refinitiv ESG scores is consistent with the signaling mechanism. Since the rating effectively functions as a signal verifier as defined by Zhang et al. (2025, p. 9), its methodology requires verifiable, externally auditable data. A CEO letter that is populated with vague commitments provides the rating agency with little substantiating evidence by Zhang et al. (2025, p. 3) theoretically argue that third party ESG ratings increase the cost of misleading by cross-referencing narrative claims against the observed company behavior. Hence, the findings of our analysis offer evidence for this mechanism in the specific context of Nordic CEO letters.

A significant aspect of the signaling interpretation concerns the identification within the sector. The signaling theory posits that equilibrium separating signals emerge within heterogeneous quality markets. High-quality firms invest in costly signals to distinguish themselves from low-quality competitors; low-quality firms rationally forgo the investment since they cannot sustain the committed risk (Spence, 1973, p. 360). The result of a negative GWRI-ESG rating relationship after the inclusion of sector fixed effects is consistent with this equilibrium

structure. In other words, the variation in letter quality is not a reflection of industry-wide rhetorical norms but of firm-level choices within the same regulatory context. Some firms write more credible sustainability communications than others, and the ESG rating agency responds to this credibility differential. Hence, the result of this study provides that CEO letters are not merely ceremonial documents whose style is dictated by the industry standard, but rather genuine strategic communication choices with observable consequences.

5.1.2 Decoupling Theory and the Temporal Structure of Greenwashing Language

As articulated by Meyer and Rowan (1977, pp. 356-359), the notion of decoupling emerged to describe the tendency of organizations in terms of adopting institutionally mandated structures and narratives whilst maintaining their technical core operations unchanged. In the sustainability context, the phenomenon of decoupling can be observed when firms publicly pledge to environmental targets and articulate ambitious visions in their CEO letters, while implementing minimal changes to their production processes, supply chains, or emission-intensive resource consumption patterns. Bernini & La Rosa (2024, p. 417) frame it as a choice of strategic disclosure, whereby the volume and positivity of environmental communication systematically exceed the level of actual performance.

The Action Temporality Ratio (C3) captures the proportion of future-oriented environmental sentences (such as the sentences that use “will”, “plan to”, “intend to) relative to the sentences that report completed or ongoing actions. Therefore, C3 is designed to detect the linguistic patterns of decoupling precisely. A textual embodiment of Meyer and Rowan’s (1977) structure occurs when a CEO letter is dominated by future commitments without evidence of current or completed actions: there is an adoption of sustainability leadership with aspirational targets, ambitious pledges without the substance of demonstrated progress, and audited outcomes. Moreover, the Commitment Horizon Distribution (C4) further captures the time depth of the decoupling. For example, a firm that commits to net zero by 2050 in its 2023 annual report does not impose significant accountability risk on its management, as the pledge horizon lies well beyond the duration of any current member of the leadership team.

The finding of our analysis that a higher composite GWRI (which mechanically reflects more commitment language for the future through C3 and C4) is associated with lower Refinitiv ESG scores indicates that decoupling in CEO letters does not go undetected by evaluators, at least not fully. This partially supports the conclusion of Barkemeyer et al. (2014, p. 254) that CEO sustainability statements tend not to impose accountability on their authors and represent primarily exercises unrelated to operational performance of the company. This study does not contradict the finding that CEO letters do not usually make the company’s leaders directly accountable to shareholders. However, it suggests that ESG rating agencies perform monitoring that shareholders alone may not. By triangulating the CEO narrative claims against the indicators of actual performance, the rating agencies can catch the differences between what the CEO says and what is really happening.

The lagged GWRI results (column 5 of Table 5) reinforce this interpretation: the CEO letter from year t-1 predicts the ESG score in year t at an even larger coefficient magnitude (-5.338 vs. -4.715). It is consistent with the view that persistent decoupling across consecutive CEO letters cumulates a larger detected gap in the annual assessment. A firm that writes temporally imbalanced CEO letters every year provides the rating agency with growing evidence that its sustainability narrative is not rooted in verifiable operational progress.

5.1.3 Impression Management and the Architecture of the CEO Letter

The study of Brennan & Merkl-Davies (2013, pp. 110-112) provides a comprehensive review of impression management strategies in corporate narrative reporting. These strategies include rhetorical manipulation, thematic emphasis, structural and visual manipulation, performance comparison selection, and attribution of outcomes. The theory explains that managers deploy these strategies to build stakeholder perceptions that exceed the reality of performance. From a sustainability perspective, Cho et al. (2015, p. 79) describe sustainability reporting as an organized hypocrisy where institutions maintain a systematic facade that serves their legitimacy. Sneideriene and Legenzova (2025, pp. 10-15) empirically demonstrate that the gap in sustainability reports is systematic and persistent rather than idiosyncratic. Additionally, the study highlights that impression management gaps are more pronounced in high stakeholder scrutiny context, which is exactly the institutional environment in which Nordic B2B firms operate.

The Structural Emphasis Score (C7) in the GWRI captures the dimension of structural manipulation as defined by Brennan and Merkl-Davies (2013, p. 118-120) which is the positioning of environmental content within the CEO letter. For instance, when a firm places the majority of sustainability claims on the early paragraphs of the letter, it signals that sustainability is a genuine priority because the readers' attention is mostly concentrated on the early paragraphs. On the other hand, if the sustainability claim is placed at the end of the letter, it is likely to be a compliance exercise rather than a genuine strategic statement. Even though C7 carries a lower weight of 20% in our analysis, its inclusion ensures the capture of structural dimension of impression management in addition to substantive and temporal dimensions. The results within the same industry show that full GWRI predicts ESG ratings even after controlling the industry context. Hence, the structural dimension of the letter independently contributes to the rating agency's assessment.

Moreover, Hu et al. (2024, pp. 1-19) provides evidence from a different angle. According to their study, lower readability of ESG disclosures is positively associated with greenwashing behavior. Although our analysis does not directly incorporate a readability measure as we excluded C2 (the readability subindicator) for preserving cross-run reproducibility, Hu et al. (2024) supports the idea that the combination of textual dimensions which GWRI captures is sufficient to detect the ESG rating penalty associated with impression management in CEO letters. To sum up, our three theoretical lenses signaling (C1), decoupling (C3,C4) and impression management (C7) provide an independent theoretical explanation for the H1 findings. When multiple theories all point to the same finding, then the results become more trustworthy than if only one theory explained it.

5.1.4 Economic Significance in Context

The results for H1 are statistically solid, but the practical size of the effect deserves an honest look. The key coefficient of -4.715 suggests that a one standard deviation increase in the GWRI-A (0.504 units) translates approximately to a 2.38 point drop in Refinitiv ESG score (around 14.6% of one standard deviation in ESG scores). Additionally, the implied difference is around 3 points across the interquartile range of the GWRI (0.649 units). To put the results in perspective within Refinitiv grading system, moving across a rating tier typically requires a shift of 5 to 10 points or more. Hence, a single year's worth of GWRI variation is very unlikely to push a firm into a different tier on its own. On the other hand, the picture of the results changes when the results are considered over a longer horizon. When a firm consistently scores high on the GWRI over three years (which is the full length of this analysis), there can be an

accumulation of an implied effect of around 9 ESG point, which is large enough to matter for ESG screened investment mandates.

Beyond that, it is important to be clear about what the GWRI does explain and not. The sector membership that is shaped by a firm's industry, regulatory environment, and reporting norms account for the vast majority of the variation in ESG scores as reflected in the R^2 increase from 0.023 to 0.112 once sector fixed effects are introduced. The GWRI result adds on top of that, but it acts as a secondary within sector contribution. This is not a weakness of the findings as it is theoretically what we would expect. Hence, what GWRI captures is a real but marginal layer of variation as one that operates within much larger constraints set by what industry a firm belongs to.

These results make sense when they are placed alongside the closest comparable study. Gorovaia and Makrominas (2025, pp. 438-441) find that NLP based greenwashing measures drawn from CSR reports are negatively associated with ESG scores among European listed companies with similar coefficient magnitudes that are reported here. This comparison is worth pausing as CSR reports are longer, more structured, and more extensively processed by rating agencies than CEO letters. These would normally lead us to expect a stronger statistical relationship for CSR based measures. Our results recover a similarly sized effect from a shorter and less standardized document suggesting that the CEO letter punches above its weight. Hence, it appears to carry concentrated, high-salience signals that compensate for its short length, a pattern that aligns with its standing as the most widely read section of the annual report (Na et al., 2020, p. 220).

5.2 Greenwashing Risk and Subsequent Stock Performance: An Informative Null

5.2.1 The Epistemology of Null Results in Empirical Finance

In order to get a deeper understanding of the main findings of H2, we need a more proper interpretation of the results. The central finding is the non-significance of the GWRI in predicting one-year ahead of stock returns. In empirical studies, a null result can usually be viewed as suspicious and informative if it, for example, comes from an insufficient or poorly defined model. However, when a null result comes from a properly described, and well-powered model that successfully rules out theoretical reasons for the lack of an observed impact, it becomes extremely informative. This research is an example of the latter characteristic and therefore guarantees a high level of confidence and validity of empirical findings. Since the study includes a sample size of 339 firm-year observations, sector, and year fixed effects as well as standard deviation clustered at the company level, the empirical model is defined with a degree of robustness that exceeds much previous research in the greenwashing literature. Furthermore, the robustness of the model suggests that the non-significant result is unlikely to be driven by omitted variable bias or insufficient control for time trends. The estimated 95% confidence interval suggests that the true effect of GWRI on returns (as reported in column (3)) is bounded between −4% and +9%, thereby ruling out large positive or negative impacts. This interval is informative, as it effectively excludes return effects smaller than −4 percentage points and larger than approximately 9 percentage points per unit of GWRI.

When we put these numbers into a more concrete economic context, these bounds correspond to roughly half of one standard deviation of yearly returns per unit of the index. The one-sigma

confidence ranges on the return effect per one-sigma move in the GWRI range from around −2.0% to +4.6% of yearly return because the GWRI itself has a standard deviation of 0.504. Although not negligible in absolute terms, such effects are relatively small economically in a market context where the yearly return standard deviation is 30.1%. While the analysis may lack sufficient power to detect very small return effects, it is effective in ruling out large and economically meaningful impacts. A further consideration is that the use of raw rather than risk-adjusted returns introduces a potential confound. Firms with higher greenwashing risk may systematically carry higher market beta, generating elevated expected returns that partially offset any negative greenwashing penalty. Without controlling for systematic risk through a factor model such as Fama-French, the null result in H2 cannot fully distinguish between the absence of a greenwashing effect and the presence of a compensating risk premium. This limitation is acknowledged in Section 3.6.2 and represents an avenue for future research using risk-adjusted return specifications.

Specifically, the findings clearly show that, in the particular context of Nordic B2B companies, the linguistic content of CEO letters does not function as a strong, reliable, or economically significant predictor of returns over a one-year horizon. The resulting "informative null" implies that narrative-level greenwashing risk does not carry a significant or consistent market cost. This absence of a significant relationship suggests that investors in this market do not incorporate ESG narratives into their financial decisions. This is important because it indicates that companies' environmental communication and financial performance may be largely decoupled, at least within certain industries.

5.2.2 The Event-Study Versus Cross-Sectional Return Architecture

Placing these results within the larger event-study literature about market responses to greenwashing provides the most direct and convincing explanation for the null result seen in the H2 analysis. In line with this perspective, a study by Dorfleitner et al. (2025, p. 1-2) offers a more nuanced view of how investors react to environmental differences, where they show that market reactions are far from monolithic. According to that study, situations involving smaller businesses, financially significant accusations, or particular compliance violations are typically the only ones that result in statistically significant negative anomalous returns. This suggests that rather than imposing an official punishment for all forms of greenwashing, the market filters its response based on the degree of infraction. Similarly, the research by Xu et al. (2025, p. 15) complements this evidence, finding that negative market reactions are most pronounced when the greenwashing evidence is concrete and when the firm previously enjoyed strong ESG credentials. On the other hand, although generally in line with this pattern, no statistically significant cumulative abnormal returns are found when Teti et al. (2024, p. 8) analyze a sample of European greenwashing announcements. This is consistent with the theory that, in the absence of specific scandal exposure, European markets do not consistently penalize narrative-level sustainability flaws.

In light of these findings, we must create a clear distinction between confirmed, publicity disclosed greenwashing events and the more latent greenwashing risk measured by the GWRI. As previously mentioned, the GWRI does not measure proven cases of misleading behaviour. Instead, it identifies subtle linguistic signals in CEO letters that may be associated with greenwashing such as vague language, uneven emphasis across time periods, or a tendency to focus on future commitments rather than present actions. In contrast to a formal disclosure or public scandal, these signals are incorporated into normal corporate communications rather than being clearly obvious as misconduct. In traditional event-study research, stock markets

typically react when new and clear information becomes publicly available and changes investors' beliefs about a firm's true sustainability performance. By contrast, the approach used in this study relies on a signal that is technically available in public documents but is not easily recognized or interpreted by most investors. Extracting meaning from such a signal requires specialized linguistic analysis that the average market participant is unlikely to apply consistently across firms. Consequently, there exists a significant information asymmetry between the analyst, who has computed the GWRI, and the broader market, which may consume the CEO's letter without the benefit of a systematic linguistic filter. This difference in information processing helps explain why the GWRI is not reflected in stock returns contemporaneously.

Interestingly, this interpretation is further supported by the results of H1, which highlights the role of institutions that exist between companies and investors. Particularly, ESG rating agencies tend to pick up the subtle signals captured by the GWRI but utilize their own structured methods and analysis tools, and they reflect this information in their ESG scores. This means that the GWRI signal is not ignored in general but rather processed by specialized organizations that are equipped to interpret it. Therefore, the lack of a significant result in H2 should not be seen as evidence that the GWRI signal is irrelevant. Instead, it shows that different parts of the financial system respond to this type of information in different ways. ESG agencies incorporate these signals into their assessments, while the broader stock market seems to react only when there are clearer or more visible events.

5.2.3 Market Efficiency and the Information Processing Hierarchy

When viewed from the perspective of the efficient markets hypothesis (EMH), the null result in the H2 analysis necessitates a more careful explanation than simply stating that markets are efficient. In theory, a semi-strong efficient market indicates that all publicly available information should be rapidly and accurately incorporated and reflected in stock prices. One could immediately assume that the GWRI's information should already be priced in because it is based on CEO letters, which are publicly available to any interested shareholder. From this angle, the EMH could appear to be supported rather than challenged by the lack of predictive power. It is worth mentioning that the EMH mainly applies to information that is not only publicly available but also easy and low-cost for investors to interpret. In contrast to standard financial data and clear news events that are easy to interpret, it can be argued that the GWRI captures a signal that is not easy and immediately visible to most investors. Although CEO letters are publicly available, extracting signals may require several analytical tools. These can include classifying sentences, combining different indicators, and comparing results across sectors and time, which can be costly and resource-intensive for most investors to apply. As a result, it is plausible that the signal remains difficult to identify and use in practice. This interpretation is consistent with Hartzmark & Sussman (2019), who show that even relatively simple ESG information tends to influence investor behavior primarily when it is presented in a clear and standardized format. By extension, the more subtle linguistic patterns captured by the GWRI may only affect investor decisions once they are translated into a more accessible and comparable metric such as ESG ratings.

Building on this reasoning, the H1 results suggest that ESG rating agencies, such as Refinitiv, may play a role in translating these signals into a usable form. From this perspective, the market's limited reaction to the GWRI could be interpreted as a case of "rational inattention," where investors rely on specialized intermediaries to process complex information. Rather than directly responding to the underlying textual signals, investors may instead react to ESG

ratings, which summarize this information in a standardized way. If this mechanism is in place, it is possible that any effect associated with the GWRI is already incorporated into stock prices through ESG ratings. This would help explain why the GWRI does not exhibit independent predictive power once ESG scores are included in the model. This is exactly what is observed in column (2) of Table 6, where the GWRI's independent influence vanishes in the presence of formal ESG metrics.

Furthermore, the consistency of the H2 null result across all model specifications, including the lagged model, suggests that delayed market reactions are unlikely to explain the absence of an effect. If investors were gradually incorporating this type of information over time, we anticipate that a significant effect would emerge in the lagged specification. In particular, the lagged GWRI coefficient would be expected to be negative and statistically significant if narrative-level greenwashing risk were priced with a predictable one- to two-year delay. However, the observed coefficient of 0.084 ($p \approx 0.17$) is statistically indistinguishable from zero, meaning the data neither support nor rule out a delayed penalization pattern. This means that the absence of a lagged effect cannot be used as evidence against delayed market reactions; it simply indicates that no such effect was detectable within this sample and time horizon. Instead, it suggests that, if a reaction exists, it may be relatively weak, dependent on specific events such as major scandals, or transmitted through channels other than stock returns. Overall, these findings indicate that while the GWRI may contain meaningful information about corporate behavior, its translation into market prices likely depends on the role of specialized intermediaries within the ESG ecosystem rather than on direct investor interpretation.

5.2.4 The B2B Relationship Channel and Horizon Mismatch

To further understand the findings of the H2 analysis, it can be worth considering the specific business dynamics in the B2B industry and its relationship channels. In contrast to B2C businesses, where buying decisions can be impulsive, B2B often relies on long term contracts and relationships with corporate clients and partners. Based on prior research by Walker & Wan (2012), greenwashing could potentially harm these relationships. They found in their study that "green talk" or purely symbolic environmental actions, is negatively associated with financial performance, whereas substantive actions do not show the same link. They see this as proof that greenwashing has actual financial costs, which eventually show up in the financial foundations of a company. However, it is worth arguing that these effects likely develop slowly over time rather than appearing immediately. For instance, if investors identify a gap between the company's environmental communication and its actual performance, actions such as terminating the contract or removing it from the supplier list can take longer time to materialize. Consequently, we assume that one year lag used in this study is likely too narrow to capture the potential effects that Walker & Wan (2012) noted, even if they exist in the background for the B2B firms in our sample. In order to examine the particular mechanism by Walker & Wan (2012) in the B2B sector, research using multi-year cumulative returns, for example 2-3-year gap between CEO letters disclosure and stock performance, would be more appropriate, which is further discussed in "future research" (section 6.4). This suggests that results of H2 in this study may not fully reflect longer-term dynamics and should be interpreted with caution.

Additionally, the importance of environmental issues may vary depending on the sector. According to Khan et al. (2016, p. 1698), ESG factors only to matter the most and produce economically significant variation when they are directly relevant to the core business model and industry For instance, whereas data privacy is the primary concern for firms in the

technology sector, while carbon emission is the crucial indicator for the energy-intensive industries, such as shipping companies or firms that produce energy etc. We assume that this variation plays a role in our results. GWRI was designed as a standardized analytical approach to CEO letters across a broad spectrum of sectors, from construction and heavy machinery to business services. Therefore, since the GWRI applies to the same measure across different sectors, it may not adequately capture these differences. For example, a construction company could potentially be exposed to significant financial risk because of strict regulatory and customer demands and expectations. In contrast, an IT firm could face little to no financial risk.

Because of the aggregate GWRI coefficient averages across these vastly different materiality environments, any sector-specific signals are likely diluted into statistical noise, which is supported by the information in column (3) in table 5. The data shows that sector fixed effects increase the R2 from 0.003 to 0.031, and notably, this increase is not a result of any significant interaction with the GWRI. Instead, it is the explanatory power of the sector classifications themselves. This indicates that while industry differences play a crucial role in explaining returns, the uniform GWRI measure lacks the precision needed to reflect the industry-specific materiality that investors tend to incorporate into their valuations. As such, this limitation in the construction of the GWRI represents one of the main constraints of our study and should be taken into account when interpreting the results.

5.2.5 ESG Momentum and the Limitations of Static Levels

The dynamic asset-pricing paradigm created by Berk et al. (2023) provides another intriguing and thorough explanation of the H2 data. This framework shows that relative change over time is more important than ESG values. While uneven ESG performance is penalized, "ESG momentum," or how quickly a company improves in comparison to peers, drives return. In the context of this study, the GWRI is examined as a "static" level taken from a CEO letter from a single year, which may unintentionally ignore important changes over time that actually matter for how markets react, as noted by Berk et al. (2023). For instance, if a firm that has a high GWRI in one period but then drastically decreases in the next period, perhaps due to internal improvements or regulatory pressure, this would for example constitute an "ESG momentum reversal" that their framework predicts would be positively priced. On the contrary, if a firm has consistently high GWRI over several years, this may signal ongoing problems, which could eventually lead to stronger negative reactions. Therefore, it is also important to focus on how it changes over time, rather only on the level of GWRI in a specific year, since any increases or decreases in the GWRI (its “momentum”) might help explain why no clear effect was found in the current H2 analysis and could better align the results with modern asset pricing theories.

Additionally, these results can be interpreted in light of the broader empirical literature, particularly the meta-analysis conducted by Friede et al. (2015), which reviews over 2,000 studies on the relationship between ESG and financial performance. The authors find that factors such as time horizon, geographic context, and measurement approaches play an important role in shaping observed outcomes, although the majority of results are non-negative. At the same time, the relationship is not consistently observed across all empirical settings, highlighting its context-dependent nature. Viewed from this perspective, our findings in H2 are broadly consistent with the existing body of evidence. We found no significant result in one year's return within a three-year Nordic B2B firm panel, which aligns well with the scenario found by Friede et al. (2015), which predicts the most uncertain or weakest results. This result contributes to the literature rather than criticizing it by offering evidence from a particular context that has not received much attention. Overall, it emphasizes the significance of taking

time dynamics into account when assessing sustainability-related financial effects and supports the idea that null results are frequent and theoretically plausible in short-horizon ESG studies, especially within geographically concentrated samples.

5.3 A Two-Audience Model of ESG Information Intermediation

The H1 analysis found a clear negative relationship between GWRI and ESG ratings, while H2 found the absence of a statistically significant relationship between GWRI and stock performance. The combined pattern can be interpreted within a larger framework related to how sustainability communication is processed across different actors in capital markets. The findings can be understood by reflecting different layers of interpretations applied by different types of market participants and not treating them independently. Interpreting the findings in combination leads to what is described as a two-audience model of ESG information intermediation. According to this paradigm, general investors and specialized intermediaries react differently and at various depths to the same underlying signal. It can be broken down into four interconnected components, which together help explain why the GWRI appears to be relevant for ESG ratings but not directly for stock returns.

The first component is related to the production of the signal itself and the way the CEOs shape the letters. CEOs typically have access to confidential information that is not entirely visible to external stakeholders regarding the sustainability of performance and future trajectory of their company. However, they have an enormous amount of freedom in how this information is presented in annual reports, especially in narrative sections like the CEO letter. Building on agency theory (Jensen and Meckling, 1976), this puts managers in a position where they can proactively influence how their company is seen to preserve credibility, control stakeholder expectations, or steer clear of possible expenses related to unfavorable disclosures. This can lead to a selective emphasis on environmentally beneficial behaviors in the context of sustainability, while less desirable features are minimized or presented in more ambiguous terms. Because narrative disclosures are less standardized and more variable than numerical financial reporting, previous research indicates that these types of greenwashing can continue even in situations that are comparatively well-regulated. The CEO letter is particularly important in this regard because it is still one of the annual report's least structured sections, offering many possibilities for impression control and narrative framing.

The second component focuses on the ESG ratings agencies, in this case on Refinitiv, which acts as a first key audience for these signals. As previously mentioned, such agencies operate as specialized intermediates between companies and investors, whose role is to assess and compare firms' sustainability performance using structured methodologies and analytical processes that may include scoring frameworks, cross-firm comparability. This process helps them to extract meaningful information that otherwise is unstructured. According to H1 results, even after accounting for sectoral and temporal factors, enterprises with higher GWRI values typically earn systematically lower ESG rankings, suggesting that ESG rating agencies are at least partially able to identify the kinds of language patterns recorded by the GWRI. This suggests that by converting intricate story signals into standardized sustainability indicators, these organizations serve as crucial filters within the information ecosystem, minimizing asymmetries between businesses and external stakeholders. By doing this, they successfully convert qualitative textual cues into numeric scores that the general public can use more readily.

The third component addresses the second audience of this study, which are the investors, whose behaviors appear to differ significantly from that of ESG ratings agencies. The results of H2 suggest that within one-year horizon examined, investors do not independently price the GWRI signal. However, this does not mean that the information in the CEO letters or the captured GWRI score is irrelevant, but it suggests that investors may not directly focus or interpret this raw information. Their financial decisions seem to be dependent on the guided and standardized indicators, such as ESG ratings, because they help them to summarize complex information in a more accessible format. Looking from this perspective, investors may respond to the ESG ratings that already incorporate, at least partially, rather than GWRI directly, if the information contains linguistics signals. If this is the case, then the effect of the GWRI may already be indirectly reflected in stock prices through ESG channels, which would explain why the GWRI loses its independent predictive power once ESG scores are included in the regression model. Our analysis of H2 can be better interpreted as an outcome of information aggregation and processing rather than as evidence that the signal itself is economical relevant. However, a more detailed analysis of the relationship between ESG ratings and stock returns is required to fully understand the extended ESG scores to transmit information into market prices.

The fourth component concerns the dynamic implications of this intermediation process. Any influence on stock prices that results from the GWRI signal being sent to investors through ESG ratings is expected to occur gradually, reflecting the pace at which capital allocation and investment decisions incorporate ESG-related information. Previous research suggests that investor reactions to ESG information often develop over time rather than instantaneously, as capital flows and portfolio adjustments require time to materialize (Krüger, 2015, pp. 304-305; Lins et al., 2017, pp. 1785-1788). This suggests that the potential effects of GWRI-related signals may only become observable over longer time horizons than those considered in the current analysis. Evidence further indicates that ESG-related performance is more likely to be reflected in financial outcomes over extended periods, particularly when sustainability issues are financially material (Khan et al., 2016, p. 1699). Therefore, a longer-term relationship between GWRI and stock returns may still exist even if no discernible impact is observed within a one-year period. Rather than rejecting the relevance of GWRI signals, this finding highlights the importance of considering the temporal dimension of information processing and market adjustment. In line with prior literature emphasizing the context-dependent nature of ESG-financial performance relationships (Friede et al., 2015, p. 211), future research should extend this approach by examining multi-year return dynamics or changes in ESG ratings over time.

Finally, the model also has important implications beyond the equity market, particularly in a B2B context where sustainability considerations increasingly influence procurement decisions. In addition to shaping investor perceptions, ESG ratings are often used by firms to evaluate potential suppliers and partners. A lower ESG rating, potentially driven by higher GWRI scores, may therefore lead to tangible economic consequences that are not immediately reflected in stock prices. These may include reduced access to contracts, stricter compliance requirements imposed by larger clients, and reputational damage within supply chains. Such effects operate through operational and strategic channels rather than through capital market valuation, and they may materialize more quickly or more strongly than any impact on stock returns. This suggests that the costs of narrative greenwashing for B2B firms may be more directly observed in revenue and margin outcomes than in short-term equity market reactions. As a result, the theoretical relevance of the GWRI may extend beyond traditional asset pricing

frameworks, pointing instead to a broader set of mechanisms through which sustainability-related information influences firm performance.

6. Conclusion

This chapter is a summary of analysis and discussion, where H1 and H2 are addressed. It also summarizes the theoretical, methodological, and practical contributions. Furthermore, it will discuss the ethical and societal implications as well as limitations of the study. Finally, the chapter will give directions for future research.

6.1 Summary of Findings and Answers to the Research Questions

The main objective of this paper was to examine the presence of greenwashing risk in CEO letters and how these risks are related to external ESG ratings and subsequent stock returns among Nordic B2B companies. To achieve this, a Greenwashing Warning Risk Index (GWRI) was built and applied to a panel of 339 CEO letter observations from a sample size of 113 firms covering the fiscal year between 2022-2024. Importantly, GWRI was created to get beyond human subjectivity and the inconsistent nature of LLM-based classifications, which frequently have reproducibility issues. Instead, GWRI creates a more transparent and repeatable language risk profile by combining four rule-based indicators that are derived from theoretical frameworks in chapter 2, including signaling theory (C1), decoupling theory (C3 and C4), and impression management theory (C7).

The first research question is focused on the relationship between the greenwashing risk embedded in CEO letters and the third party ESG ratings: *How does the greenwashing risk in executive messages relate to third-party ESG ratings of B2B Nordic firms?* From this point, the empirical data is clear and shows that greenwashing risk measured by GWRI and the ESG scores have a strong and negative correlation, which were well supported by all the four robustness tests as part of the study's design. Based on the main model that accounts for differences across firms, years and industries, a one-unit increase in GWRI-A is linked to a 4.72-point decrease in a firm's ESG score. To put this measurement into context, if a company moves from a relatively low to a relatively high level of GWRI (from the 25th to the 75th percentile of the distribution), it corresponds to an ESG score reduction by roughly 3.1-point drop, considering that all other variables are held constant. Furthermore, the outcome is nearly identical when the data was adjusted to reduce the impact of extreme values (winsorization). The decrease is considerably more pronounced when the study utilizes GWRI values (lagged by one year). These tests are crucial because they demonstrate that the outcome is not influenced by atypical extreme cases by subsequently changing their language. In other words, it reduces concerns and reverses the causality of outliers. In summary, the initial hypothesis H1 is strongly supported because the outcome remains consistent across all of these tests, and there is a strong correlation between greenwashing risk and ESG ratings. From a theoretical standpoint, these results can be comprehended by three related mechanisms. Firstly, ESG ratings agencies act as "signal verifiers", penalizing CEO letters that contain vague or unsupported environmental statements, consistent with Spence's (1973), signaling theory. Second, the results may indicate "organizational decoupling" where firms rely heavily on future promises where stated actions differ from actual practices. ESG ratings agencies in turn, are able to detect this gap and reflect their scores. Finally, how sustainability information is structure relates to impression management which suggests that not only content but also the presentation of the information influences perceived corporate transparency.

The main objective of the second research question is to examine the relationship between the measured greenwashing risk and the subsequent stock performance: *How do these executive sustainability messages, and any associated greenwashing risk, affect subsequent stock market*

performance? Here, the empirical evidence shows the opposite of H1, thus, there is no clear correlation between GWRI and stock market performance. The estimates are very close to zero and not statistically significant. Notably, the ESG scores itself failed to demonstrate any predictions for stock returns in this specific sample. Consequently, H2 is not supported. This non-result, however, should not be considered a failure of the empirical design. Instead, as discussed in chapter 5, it is an informative discovery, suggesting that rather than being independently or directly priced by investors, the rhetorical signals about sustainability are filtered through the ESG grading system at a one-year horizon. Rather than immediate change in stock prices, any consequence of greenwashing is likely to appear more slowly or through other channels, such as changes in investment decisions or business partnerships.

In summary, these results point to a two-audience model in how sustainability information is interpreted, as previously mentioned. In this model, ESG rating agencies, as specialized actors, have the ability to carefully analyze the language of corporate disclosures and are capable of detecting the risk of greenwashing. On the other hand, investors do not seem to detect these risks directly, but they heavily rely on ESG ratings that incorporate them. In simple terms, investors focus on more accessible information rather than detailed linguistic analysis, suggesting that the financial effects of greenwashing are likely to unfold indirectly or over a longer time horizon.

6.2 Contributions

6.2.1 Theoretical Contributions

This study involves three distinct streams of academic literature, and it is worth being clear about what each contribution actually adds. The first and most clear contribution is to the empirical literature on the CEO letters as a corporate communication instrument. Arvidsson (2023, pp. 50-53) reports the growing prevalence of sustainability narratives in Nordic CEO letters and warns that the increasing sophistication of the letters does not necessarily reflect genuine sustainability embeddedness. Our study takes the qualitative warning of Arvidsson and turns it into a testable study. As far as the authors are aware, our study provides the first panel-regression evidence that the linguistic quality of CEO letter sustainability content is systematically evaluated with third party ESG assessments within the Nordic B2B segment. It moves the existing literature beyond describing and interpreting what CEO letters try to convey, toward actually identifying what those choices actually produce. This shift matters since it gives empirical grounding to a claim that has long been made on theoretical terms: that CEO letter rhetoric is not merely symbolic, but economically consequential.

The second contribution sits within the theoretical and empirical literature on greenwashing and its market consequences. Bernini and La Rosa (2024, p. 425) identify CEO letter specific greenwashing metrics as a critical gap, and our analysis with GWRI speaks directly to that identified gap. Each GWRI sub-indicator is grounded in a distinct and established theoretical framework. Signaling theory underpins C1, decoupling theory underpins C3 and C4, and impression management theory underpins C7. The GWRI also empirically demonstrates that the composite instrument detects a meaningful and robust signal in a well identified panel setting.

The next contribution is to the two-audience model of ESG information intermediation. Zhang et al. (2025, p. 9) argue that rating agencies function as information intermediaries and it is

indirectly supported by capital flow evidence of Hartzmark and Sussman (2019, pp. 2791-2792). The present study adds the first direct empirical evidence of how these two audiences differ in a controlled panel setting with a clear pattern. The H1 result shows that the audience, specifically specialized ones like the rating agency, detects and prices the signal embedded in the CEO letter, and that signal is reflected in the ESG score they assign. In a scenario where equity investors were reading CEO letters and penalizing companies for burying or vagueing up sustainability claims, we would observe a reflection in stock price movements. However, the results show no significant influence on stock price, at least not within a one-year horizon. The results imply that investors delegate this kind of detailed linguistic analysis to rating agencies rather than performing it themselves, although the non-result alone cannot rule out alternative explanations such as the signal being too weak or the horizon being too short to detect a market reaction.

6.2.2 Methodological Contribution

From a methodological standpoint, this study contributes by showing that a positivist and deductive research design can be carried through consistently from philosophy to measurement in the context of corporate sustainability disclosure. It is worth noting that this consistency was not incidental. Each methodological layer from the epistemological stance to the choice of estimator was shaped by what preceded it rather than selected in isolation. Most concretely, the development of the GWRI illustrates that when no existing tool is sufficient, a purposely built index can follow directly from the theoretical commitments of the study instead of being introduced as a separate methodological layer. On this basis, the overall design offers a transferable template for future research working at the intersection of textual analysis and financial market outcomes.

6.2.3 Practical Contributions

The study's findings, particularly H1, deliver a clear message to corporate managers and investors. The way sustainability is expressed in the CEO letters has measurable consequences for the ESG ratings a firm receives from third party agencies. This puts into questions the implicit presumption that CEO letters serve primarily as external communication tools whose content does not feed back into formal evaluation systems, which is occasionally present in practice. The empirical results indicate that, while keeping actual operational performance constant, increasing the specificality of environmental statements (moving from lower scores of 0 or 1 to higher scores of 3 or 4 on the C1 rubric), changing the temporal focus of environmental disclosures toward realized actions rather than future commitments (reducing C3), and making sure that forward-looking statements are grounded in near-term rather than distant promises (reducing C4) are all associated with higher ESG ratings. This emphasis that the way sustainability performance is disclosure without significant changes in performance. Therefore, CEOs and sustainability officers should be aware that the same underlying acts might be presented in ways that range significantly in terms of credibility and clarity, and that ESG rating agencies seem to react to theses variations in narrative structure in a methodological manner.

The combined results point to a complimentary interpretation for ESG analysts and equities investors. In the Nordic B2B setting, the GWRI does not seem to produce short-horizon trading signals, but it does seem to contain diagnostic data regarding companies whose sustainability story is not entirely consistent with their underlying performance. A GWRI-based analysis of CEO letters may therefore serve as an additional tool for identifying companies at risk of future ESG rating adjustments when used in conjunction with Refinitiv ESG scores rather than in

place of them. This is especially true when companies exhibit high GWRI values while their ESG scores are near categorical thresholds, where even slight downgrades could have significant portfolio implications due to ESG screening rules. This also suggests that exposure to companies with high GWRI values may carry forms of supply-chain or counterparty risk that are not fully represented in current ESG scores for institutional investors operating under sustainability-linked mandates or procurement limits. The GWRI can therefore provide pertinent information about fundamental sustainability mismatch that may become economically significant through other institutional channels, even though it may not directly convert into immediate return predictability.

6.3 Ethical and Societal Implications

Conducting research ethically is not only about how the data is handled. It also involves being honest about what the findings mean for people outside of academia (Saunders et al., 2019, pp. 140-150). In this study, this question is particularly relevant. Firms that design CEO letters signaling stronger sustainability credentials than their actual ESG performance warrants are creating an information gap, and that gap does not land equally across all investor types. Retail investors, who rarely have access to the kind of analytical infrastructure that institutional investors rely on tend to take narrative disclosures at face value more often. For this particular reason, CEO letters are not a neutral communication channel. They are one of the few touchpoints through which smaller investors form impressions of a firm´s sustainability profile, which makes the rhetorical patterns this study identifies more than an academic concern. A firm that scores high on the GWRI is a firm whose sustainability narrative may be misleading the very investors least equipped to question in practical terms. These greenwashing concerns are becoming more significant as EU regulations give more importance to it recently. For instance, The EU Green Claims Directive (2023) makes it mandatory for companies to prove their environmental claims are true before publishing them. The GWRI that was created for this research is exactly in line with what the directive is trying to eliminate. Additionally, European companies are forced to report sustainability in a standardized way by CSRD. The aim of the standardization is to remove the freedom of firms that can be used as manipulation such as by giving vague commitment claims, using heavily future oriented language without current commitment and burying crucial environmental content. Nordic B2B firms are specifically important in this context since the Nordic region is one of the first regions where CSRD applies. Hence, the selected firms for this research are directly affected by these rules. Therefore, the findings of the analysis are not only academically relevant but also relevant in terms of recent policies, and they represent exactly what these directives are designed to prevent.

6.4 Limitations

The biggest limitation of the research is related to what GWRI can capture. The tool is limited to capturing greenwashing risk, and it does not guarantee greenwashing at hundred percent. Since the index measures rhetorical patterns associated with greenwashing and not actual deceptive intent. There can be cases where a firm could score high due to genuinely uncertain future commitments rather than deliberate manipulation. Therefore, it limits how strongly H1 findings can be interpreted.

Additionally, H2 findings deserve careful interpretation as there is a statistically insignificant relationship between GWRI and forward stock returns does not necessarily mean that greenwashing risk is irrelevant to financial markets. There may be a plausible explanation that

lies in the design of the test itself. Stock markets may tend to react to greenwashing upon discrete, observable events such as regulatory investigations, investigative journalism, NGO reports rather than in response to annual CEO letters alone. Hence, a one-year stock return window is unlikely to capture this kind of reaction. The embedded signal in a CEO letter might take longer than twelve months to be recognized, verified, and priced by the market participants. Considering this, the null result for H2 is better understood as a limitation of the measurement horizon rather than evidence against the underlying hypothesis.

Thirdly, using only a single ESG rating provider means the findings are conditional on chosen provider´s specific methodology and the results may not hold under other providers like MSCI and Sustainalytics. Moreover, absence of financial controls like firms' size, leverage, and profitability means residual confounding cannot be fully ruled out. There is also a selection bias toward larger, more internationally oriented firms due to the selection of English language letters only. Hence, smaller Nordic firms that are published only in local languages are excluded.

It is also significant to consider that the chosen years 2022-2024 are in an atypical macroeconomic period that includes the energy crisis, post-Covid recovery, and aggressive monetary tightening. Likely, these macro shocks dominated any greenwashing signal in stock returns, which partially explains the null results of H2. Beyond this, the research only focuses on one communication channel (CEO letters) out of many that firms use. Firms communicate sustainability through annual reports, sustainability reports, and press releases. Focusing only on CEO letters may miss compensatory or contradictory signals in other channels. Last point worth raising is that the chosen theory-based composite weights are not empirically validated. The 40/40/20 weighting of GWRI-A does not reflect data-driven optimization but theoretical reasoning. Neither of the weight structure is formally tested against an external criterion even though GWRI-B provides a sensitivity check.

6.5 Directions for Future Research

Based on the results of this study, we have identified several avenues for future research. To begin with, future research could examine whether the relationship between the GWRI and ESG ratings remains consistent across different providers, such as MSCI, Sustainalytics, and ISS. Such a comparison would clarify the generalizability of the results and show it some providers are more sensitive to the language subtleties of greenwashing than others, given the well-documented divergence in ESG grading techniques. Moreover, as previously mentioned in chapter 5, a future study could investigate whether the financial consequences of greenwashing emerge over a longer time horizon. This study did not find any statistically significant relationship between greenwashing and stock return over one-year horizon, however; applying event-study approaches or multi-year stock returns could provide a deeper insight into how markets finally react to hazards associated with greenwashing risk.

Furthermore, the geographic and sectoral score of the analysis could be expanded beyond Nordic B2B firms. The GWRI framework should be tested in other European markets or in US-based samples as well as other sectors like B2C firms, to ascertain if the results are unique to the Nordic institutional environment or generalizable across different market contexts. Finally, it would be interesting to examine alternative outcome variables beyond stock returns. Since the evidence indicates that greenwashing may impact business through indirect channels, including other variables such as supplier retention, contract renewal or revenues may offer a

more accurate understanding of how sustainability credibility affects a firm's long term commercial performance.

Overall, the methodological and analytical framework developed in this research offer a solid basis for future studies on sustainability communication and greenwashing risk. By expanding the GWRI framework across various markets, ESG ratings methodologies, and longer-term performance measures, future studies can deepen the understanding of how corporate sustainability narratives are interpreted and how their effects manifest in financial and business environment.

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Appendices

Appendix A - Dataset Variable Definitions

This appendix provides variable-level documentation for each of the three datasets used in the empirical analysis. Variables are listed with their data type, analytical role, a brief description, and their original source.

A1. GWRI Dataset (Greenwashing Warning Risk Index)

Contains 381 firm-year observations across 127 firms (the raw GWRI dataset prior to final sample construction). After removing 14 firms for which stock return data was unavailable, the final estimation sample is reduced to 339 firm-year observations across 113 firms, as described in Section 3.4.3.

Table A1: GWRI Dataset

Variable	Type	Role	Description	Source
ID	Integer	Unique	Row identifier	Constructed
document_id	String	Unique	Company name + report year (slug format)	Constructed
company	String	Panel	Full legal company name	Refinitiv/LSEG
ticker	String	Panel	Stock exchange ticker symbol	Refinitiv/LSEG
year	Integer	Panel	Fiscal year of the annual report	Refinitiv/LSEG
industry	String	Panel	Industry classification of the company	Refinitiv/LSEG
n_sentences_total	Integer	Obs.	Total sentence count in annual report sustainability section	Constructed (NLP)
n_environmental	Integer	Obs.	Count of sentences classified as environmental in content	Constructed (NLP)
gwri_version_a	Float	Key var.	Greenwashing Warning Risk Index - Version A (theory-weighted: 40% Substantive Disclosure Quality, 40% Commitment Accountability, 20% Rhetorical Positioning)	Constructed (NLP)
gwri_version_b	Float	Key var.	Greenwashing Warning Risk Index - Version B (equal-weighted: 25% to each of C1, C3, C4, C7)	Constructed (NLP)

A2. ESG Score Dataset

Contains 414 firm-year observations with ESG scores sourced from Refinitiv (LSEG). The ESG score is a composite measure (0-100) reflecting environmental, social, and governance performance.

Table A2: ESG Score Dataset

Variable	Type	Role	Description	Source
company	String	Panel	Full legal company name	Refinitiv/LSEG

ticker	String	Panel	Stock exchange ticker symbol	Refinitiv/LSEG
year	Integer	Panel	Fiscal year of observation	Refinitiv/LSEG
esg_score	Float	Key var.	Combined ESG score (0-100 scale), sourced from Refinitiv ESG database	Refinitiv/LSEG

A3. Stock Return Dataset

Contains year-end closing prices for 138 Nordic-listed companies over 2022-2025. Annual returns were computed from consecutive year-end prices and used as the dependent variable in regression analyses.

Table A3: Stock Return Dataset

Variable	Type	Role	Description	Source
ticker	String	Panel	Stock exchange ticker symbol	Refinitiv/LSEG
2022	Float	Obs.	Closing stock price at end of year 2022 (local currency)	Refinitiv/LSEG
2023	Float	Obs.	Closing stock price at end of year 2023 (local currency)	Refinitiv/LSEG
2024	Float	Obs.	Closing stock price at end of year 2024 (local currency)	Refinitiv/LSEG
2025	Float	Obs.	Closing stock price at end of year 2025 (local currency)	Refinitiv/LSEG
return_2023	Float	Key var.	Annual stock return for 2023: (P2023 - P2022) / P2022	Constructed
return_2024	Float	Key var.	Annual stock return for 2024: (P2024 - P2023) / P2023	Constructed
return_2025	Float	Key var.	Annual stock return for 2025: (P2025 - P2024) / P2024	Constructed

Appendix B - Full Stock Return Data

Year-end closing prices (in local currency) and computed annual returns for all 138 Nordic-listed companies included in the sample. Returns are expressed as proportional changes: Return_t = (Price_t - Price_{t-1}) / Price_{t-1}.

Table B1: Full Stock Return Data

Ticker	Price 2022	Price 2023	Price 2024	Price 2025	Return 2023	Return 2024	Return 2025
EQNR.OL	351.80	322.15	265.40	237.00	-8.43%	-17.62%	-10.70%
KOG.OL	82.41	92.28	253.81	256.39	11.98%	175.03%	1.02%
NHY.OL	73.32	68.40	62.54	78.20	-6.71%	-8.57%	25.04%
YAR.OL	430.60	361.20	300.80	414.00	-16.12%	-16.72%	37.63%
VAR.OL	33.60	32.16	35.29	33.00	-4.30%	9.73%	-6.49%
SUBC.OL	113.05	148.20	180.10	203.20	31.09%	21.52%	12.83%
AKER.OL	719.50	666.00	549.00	768.00	-7.44%	-17.57%	39.89%
WAWI.OL	90.76	83.23	87.44	97.61	-8.29%	5.06%	11.62%
TOM.OL	165.60	123.45	146.60	136.00	-25.45%	18.75%	-7.23%
AUTO.OL	17.93	19.98	11.11	11.81	11.46%	-44.39%	6.30%
WWI.OL	274.00	364.50	409.50	590.00	33.03%	12.35%	44.08%
NOD.OL	163.90	126.00	100.45	133.30	-23.12%	-20.28%	32.70%

TGS.OL	132.00	132.20	113.40	91.65	0.15%	-14.22%	-19.18%
VEI.OL	96.80	102.20	142.20	178.60	5.58%	39.14%	25.60%
ODLO.OL	26.30	39.40	51.30	87.50	49.81%	30.20%	70.57%
AKSOA.OL	33.46	42.02	31.10	30.92	25.57%	-25.99%	-0.58%
CADLR.OL	38.40	46.92	63.70	48.04	22.19%	35.76%	-24.58%
SNI.OL	269.50	311.50	289.00	326.00	15.58%	-7.22%	12.80%
ELK.OL	35.20	21.16	17.52	30.48	-39.89%	-17.20%	73.97%
BRGB.OL	152.00	171.40	182.40	199.40	12.76%	6.42%	9.32%
ATEA.OL	114.00	129.20	141.40	157.80	13.33%	9.44%	11.60%
BWO.OL	23.78	22.42	29.95	45.20	-5.72%	33.59%	50.92%
ODF.OL	89.00	116.50	116.80	127.00	30.90%	0.26%	8.73%
AKBM.OL	38.05	46.00	67.10	98.70	20.89%	45.87%	47.09%
MULTI.OL	139.00	124.00	199.00	177.00	-10.79%	60.48%	-11.06%
NEL.OL	12.78	6.36	2.70	2.23	-50.27%	-57.46%	-17.46%
BEWI.OL	45.90	25.70	26.60	18.76	-44.01%	3.50%	-29.47%
HEX.OL	20.97	28.88	45.50	7.74	37.69%	57.55%	-82.99%
ACR.OL	5.88	5.08	3.69	7.78	-13.61%	-27.36%	110.84%
KOA.OL	2.52	2.05	1.93	2.08	-18.65%	-5.66%	7.29%
ATCOa.ST	123.10	173.55	168.85	166.05	40.98%	-2.71%	-1.66%
VOLVaSEK.xbo	198.10	267.00	270.20	296.40	34.78%	1.20%	9.70%
SAND.ST	188.40	218.10	198.30	300.60	15.76%	-9.08%	51.59%
ERICbSEK.xbo	60.90	63.11	89.88	90.60	3.63%	42.42%	0.80%
ASSAb.ST	223.70	290.30	326.80	358.90	29.77%	12.57%	9.82%
SAABb.ST	102.67	151.90	233.70	537.60	47.94%	53.85%	130.04%
EPIRb.ST	167.65	176.40	172.40	186.70	5.22%	-2.27%	8.29%
HPUR.OL	20.70	11.14	5.60	1.43	-46.18%	-49.73%	-74.46%
ALFA.ST	301.10	403.40	462.60	465.70	33.98%	14.68%	0.67%
BOL.ST	379.12	314.50	310.50	515.20	-17.04%	-1.27%	65.93%
SKFb.ST	159.15	201.30	207.60	245.80	26.48%	3.13%	18.40%
SKAb.ST	165.00	182.35	232.70	252.30	10.52%	27.61%	8.42%
SECUb.ST	86.96	98.58	136.90	147.20	13.36%	38.87%	7.52%
ADDTb.ST	148.70	221.40	301.20	327.40	48.89%	36.04%	8.70%
INDT.ST	211.10	261.80	277.00	240.40	24.02%	5.81%	-13.21%
SSABa.ST	56.94	76.68	44.97	70.48	34.67%	-41.35%	56.73%
TRELb.ST	240.80	337.60	378.60	392.70	40.20%	12.14%	3.72%
NIBEb.ST	97.10	70.80	43.24	35.62	-27.09%	-38.93%	-17.62%
BEIJb.ST	134.69	134.90	163.10	149.10	0.15%	20.90%	-8.58%
HOLMb.ST	413.90	425.70	406.20	354.40	2.85%	-4.58%	-12.75%
SWECb.ST	99.85	135.10	164.70	150.90	35.30%	21.91%	-8.38%
GETIb.ST	216.30	224.30	181.55	218.80	3.70%	-19.06%	20.52%
MYCR.ST	97.90	143.70	199.70	223.25	46.78%	38.97%	11.79%
SECTb.ST	146.97	179.56	276.65	251.00	22.18%	54.07%	-9.27%
LAGRb.ST	103.30	135.10	207.60	212.80	30.78%	53.66%	2.50%
MTRS.ST	102.90	163.50	186.00	171.90	58.89%	13.76%	-7.58%

LOOMIS.ST	286.00	267.60	336.40	390.40	-6.43%	25.71%	16.05%
PEABb.ST	59.10	57.10	79.20	85.35	-3.38%	38.70%	7.77%
HPOLb.ST	111.10	122.00	102.80	87.95	9.81%	-15.74%	-14.45%
HMSN.ST	339.60	497.40	433.80	419.00	46.47%	-12.79%	-3.41%
EKTAb.ST	62.80	82.32	61.15	56.75	31.08%	-25.72%	-7.20%
BRAV.ST	111.40	81.05	80.10	89.75	-27.24%	-1.17%	12.05%
NCCG.L	200.00	128.80	150.60	139.20	-35.60%	16.93%	-7.57%
BUFAB.ST	46.80	77.28	87.84	100.56	65.13%	13.66%	14.48%
BILL.ST	127.10	102.40	101.40	93.90	-19.43%	-0.98%	-7.40%
AQ.ST	61.90	101.40	140.00	197.20	63.81%	38.07%	40.86%
ALIFb.ST	108.60	109.40	137.30	159.10	0.74%	25.50%	15.88%
GRANG.ST	85.30	116.00	131.70	145.70	35.99%	13.53%	10.63%
SYSR.ST	71.80	78.70	88.70	87.10	9.61%	12.71%	-1.80%
EPROb.ST	43.82	55.00	68.50	65.20	25.51%	24.55%	-4.82%
OEMb.ST	72.30	105.40	111.80	145.40	45.78%	6.07%	30.05%
AFRY.ST	170.90	139.70	153.70	150.00	-18.26%	10.02%	-2.41%
BEIAb.ST	163.40	190.00	164.20	297.50	16.28%	-13.58%	81.18%
VITRSEK.xbo	186.20	194.70	215.00	137.00	4.56%	10.43%	-36.28%
LIABSEK.xbo	127.40	199.10	229.20	208.40	56.28%	15.12%	-9.08%
ALIGSEK.xbo	55.69	82.10	117.40	145.20	47.41%	43.00%	23.68%
NOLAbSEK.xbo	54.65	52.90	54.20	61.80	-3.20%	2.46%	14.02%
IVSO.ST	164.60	195.60	275.50	263.00	18.83%	40.85%	-4.54%
HANZA.ST	51.10	85.30	76.70	127.20	66.93%	-10.08%	65.84%
MAERSKb.CO	15030.28	11681.67	11905.00	14640.00	-22.28%	1.91%	22.97%
VWS.CO	202.10	214.30	98.08	173.40	6.04%	-54.23%	76.79%
NSISb.CO	351.90	371.10	407.70	407.80	5.46%	9.86%	0.02%
GMAB.CO	2941.00	2155.00	1492.50	2027.00	-26.73%	-30.74%	35.81%
COLOb.CO	812.00	772.00	786.20	546.40	-4.93%	1.84%	-30.50%
NKT.CO	359.37	463.80	514.50	798.50	29.06%	10.93%	55.20%
ISS.CO	147.30	128.95	131.30	217.20	-12.46%	1.82%	65.42%
DEMANT.CO	192.55	296.00	264.20	215.20	53.73%	-10.74%	-18.55%
ROCKb.CO	163.30	197.60	255.20	225.40	21.00%	29.15%	-11.68%
FLS.CO	251.70	287.20	356.00	445.00	14.10%	23.96%	25.00%
TRMDa.CO	198.40	204.20	138.40	126.90	2.92%	-32.22%	-8.31%
NETCG.CO	294.20	225.60	339.00	358.00	-23.32%	50.27%	5.60%
SCHO.CO	524.00	553.00	538.00	659.00	5.53%	-2.71%	22.49%
AMBUb.CO	89.00	105.20	104.00	88.00	18.20%	-1.14%	-15.38%
GN.CO	159.75	171.80	133.75	106.75	7.54%	-22.15%	-20.19%
PAALb.CO	262.00	322.50	502.00	874.00	23.09%	55.66%	74.10%
DNORD.CO	418.00	321.00	212.40	252.00	-23.21%	-33.83%	18.64%
CHEMM.CO	690.50	388.00	486.20	687.50	-43.81%	25.31%	41.40%
SPGP.CO	256.50	218.50	307.50	347.00	-14.81%	40.73%	12.85%
NLFSK.CO	146.60	118.20	104.40	138.80	-19.37%	-11.68%	32.95%
NTGNT.CO	238.50	294.00	256.50	189.40	23.27%	-12.76%	-26.16%

MTHH.CO	145.50	123.50	289.00	520.00	-15.12%	134.01%	79.93%
AOJb.CO	83.11	70.30	78.60	94.40	-15.41%	11.81%	20.10%
TRIFOR.CO	143.06	104.31	74.50	90.80	-27.08%	-28.58%	21.88%
SOLARb.CO	620.00	465.00	299.50	202.00	-25.00%	-35.59%	-32.55%
NNIT.CO	65.70	84.10	92.50	49.80	28.01%	9.99%	-46.16%
RTX	100.92	84.14	115.72	183.40	-16.63%	37.53%	58.49%
FLUGb.CO	425.00	326.00	332.00	318.00	-23.29%	1.84%	-4.22%
TCM.CO	73.00	45.50	67.00	68.00	-37.67%	47.25%	1.49%
NOKIA.HE	4.33	3.05	4.27	5.57	-29.47%	40.06%	30.35%
NESTE.HE	43.02	32.21	12.13	19.41	-25.13%	-62.36%	60.08%
WRT1V.HE	7.87	13.13	17.11	30.40	66.81%	30.36%	77.67%
UPM.HE	34.93	34.06	26.56	24.79	-2.49%	-22.02%	-6.66%
STERV.HE	13.15	12.53	9.72	10.71	-4.75%	-22.41%	10.16%
KCRA.HE	9.59	13.59	20.40	31.30	41.79%	50.07%	53.43%
VALMT.HE	25.16	26.11	23.33	28.33	3.78%	-10.65%	21.43%
HUH1V.HE	32.00	36.73	34.18	29.74	14.78%	-6.94%	-12.99%
KEMIRA.HE	14.33	16.79	19.52	19.58	17.17%	16.26%	0.31%
HIAB.HE	24.00	30.61	49.46	49.52	27.54%	61.62%	0.11%
OUT1V.HE	4.73	4.48	2.90	4.48	-5.22%	-35.21%	54.22%
TIETO.HE	26.52	21.54	17.02	18.30	-18.78%	-20.98%	7.52%
VAIAS.HE	39.45	39.70	48.40	44.05	0.63%	21.91%	-8.99%
TYRES.HE	9.58	8.26	7.35	9.46	-13.80%	-11.04%	28.85%
METSB.HE	8.77	7.19	4.24	3.11	-17.97%	-41.00%	-26.64%
SCANFL.HE	6.58	7.83	8.25	9.98	19.00%	5.36%	20.97%
PON1V.HE	25.30	22.60	20.10	25.90	-10.67%	-11.06%	28.86%
OLVAS.HE	33.15	28.05	29.20	31.35	-15.38%	4.10%	7.36%
QTCOM.HE	44.50	64.50	67.20	32.94	44.94%	4.19%	-50.98%
ENENTO.HE	20.14	19.48	17.48	15.86	-3.30%	-10.27%	-9.27%
ICP1V.HE	17.10	7.75	10.24	9.82	-54.68%	32.13%	-4.10%
GOFORE.HE	22.20	22.40	22.20	13.48	0.90%	-0.89%	-39.28%
ETTE.HE	14.60	13.80	10.00	9.48	-5.48%	-27.54%	-5.20%
ASPO.HE	8.20	5.98	4.85	6.52	-27.07%	-18.90%	34.43%
RAUTE.HE	7.82	9.84	13.10	15.10	25.83%	33.13%	15.27%
SUY1V.HE	3.00	2.85	2.28	1.79	-4.84%	-20.00%	-21.49%
TLT1V.HE	3.54	2.70	2.64	3.76	-23.73%	-2.22%	42.42%
TNOM.HE	6.09	4.15	2.72	2.02	-31.87%	-34.52%	-25.49%
EXELC.HE	19.29	9.04	4.20	6.76	-53.14%	-53.55%	61.07%
SUMMA.HE	7.04	4.90	3.44	2.12	-30.40%	-29.80%	-38.37%

Appendix C - GWRI Construction Code

This appendix presents the essential Python code for constructing the Greenwashing Warning Risk Index (GWRI). Four files are shown: the two configuration files that define all scoring constants and lexicons, the core variable-scoring functions (C1, C3, C4, C7), and the composite aggregation logic. Language: Python 3.13. Full source including preprocessing, segmentation,

AI-assisted variables (C2, C5, C6), and validation modules is available from the authors upon request.

C.1 Environmental and Buzzword Lexicons (config/lexicons.py)

Defines ENVIRONMENTAL_KEYWORDS, used to identify environmental sentences (Module 3), and BUZZWORD_LEXICON, used in the C2 Lexical Inflation Risk calculation.

```
# ==============================================================================
# config/lexicons.py
# ==============================================================================
# Two curated word lists that drive environmental sentence detection (Module 3)
# and the Lexical Inflation Risk variable (C2).
#
# ENVIRONMENTAL_KEYWORDS  - terms that signal environmental content in a sentence
# BUZZWORD_LEXICON        - green marketing terms used in the C2 interaction score
#
# Both lists use lowercase only. Matching is done case-insensitively.
# To add or remove a term, edit the list here. No other file needs changing.
# ==============================================================================


# --- Environmental Keyword Lexicon ------------------------------------------
# Used by Module 3 to identify environmental sentences.
# Sources: GHG Protocol terminology, TCFD framework, academic greenwashing
# literature (e.g., Lyon & Maxwell 2011; Delmas & Burbano 2011).


ENVIRONMENTAL_KEYWORDS = [


    # Climate and atmosphere
    "climate",
    "climate change",
    "climate risk",
    "global warming",
    "temperature",
    "atmosphere",
    "atmospheric",


    # Carbon and emissions
    "carbon",
    "carbon dioxide",
    "co2",
    "emissions",
    "greenhouse",
    "ghg",
    "greenhouse gas",
    "scope 1",
    "scope 2",
    "scope 3",
    "decarbonisation",
    "decarbonization",
    "carbon footprint",
    "carbon neutral",
    "carbon neutrality",
    "net zero",
    "net-zero",
    "zero carbon",
    "low carbon",
    "carbon offset",
    "carbon capture",
    "carbon credit",
```

```
    # Energy
    "renewable",
    "renewable energy",
    "solar",
    "wind energy",
    "wind power",
    "hydropower",
    "geothermal",
    "clean energy",
    "energy transition",
    "energy efficiency",
    "fossil fuel",
    "fossil fuels",
    "coal",
    "natural gas",
    "electricity consumption",
    "energy consumption",
    "energy use",
    "kilowatt",
    "megawatt",


    # Nature and biodiversity
    "biodiversity",
    "ecosystem",
    "deforestation",
    "reforestation",
    "afforestation",
    "forest",
    "wildlife",
    "habitat",
    "nature",
    "land use",
    "land degradation",
    "ocean",
    "marine",


    # Water
    "water usage",
    "water consumption",
    "water stewardship",
    "water efficiency",
    "water scarcity",
    "wastewater",


    # Waste and circular economy
    "waste",
    "recycling",
    "circular economy",
    "landfill",
    "single-use plastic",
    "plastic",
    "packaging",


    # Sustainability (general)
    "sustainability",
    "sustainable",
    "sustainable development",
    "environmental",
    "environment",
    "ecological",
    "ecology",
    "planet",
```

```
    "green",
    "esg",
    "responsible sourcing",
    "supply chain emissions",
    "paris agreement",
    "cop",
    "ipcc",
    "tcfd",
    "gri reporting",
    "science-based targets",
    "sbti",
    "iso 14001",
]


# --- Buzzword Lexicon ----------------------------------------------------------
# Used in the C2 (Lexical Inflation Risk) calculation.
# These are green marketing terms that are typically vague, aspirational, or
# brand-identity-oriented rather than substantive.
# A high density of these terms combined with low claim specificity (C1) is
# the C2 risk signal.
#
# Source rationale: terms frequently identified in greenwashing communication
# research as impressionistic rather than informational.


BUZZWORD_LEXICON = [


    # Identity and aspiration
    "sustainability leader",
    "sustainability champion",
    "sustainability journey",
    "sustainability roadmap",
    "sustainability agenda",
    "responsible business",
    "responsible company",
    "purpose-driven",
    "purpose driven",
    "values-driven",


    # Vague positive framing
    "better world",
    "sustainable world",
    "greener future",
    "sustainable future",
    "green future",
    "planet-positive",
    "planet positive",
    "future generations",
    "generations to come",
    "people and planet",
    "planet first",


    # Emotive commitment language
    "passionate about",
    "deeply committed",
    "proud to be",
    "proud to lead",
    "proud of our progress",
    "committed to the planet",
    "committed to a sustainable",
    "making a difference",
    "positive impact",
```

```
    "meaningful impact",
    "real change",
    "bold commitment",
    "strong commitment",
    "ambitious commitment",
    "ambitious target",


    # Vague descriptors
    "eco-friendly",
    "eco friendly",
    "environmentally friendly",
    "green products",
    "green solutions",
    "responsible",
    "holistic approach",
    "integrated approach",
    "transformative",
    "pioneering",
    "industry-leading sustainability",
    "world-class sustainability",


    # Action theatre
    "sustainability in everything we do",
    "embedded in our dna",
    "at the heart of everything",
    "woven into our strategy",
    "change maker",
    "driving change",
    "leading the way",
    "ahead of the curve",
]
```

C.2 Scoring Rules and Weights (config/rules.py)

Single source of truth for all research-design constants: the C1 specificity rubric (0-4), C4 temporal horizon classification rules, subdimension structure and weights for Version A (theory-based) and Version B (equal-weighted), and winsorization percentiles.

```
# ============================================================================
# config/rules.py
# ============================================================================
# All theory-driven scoring rules, weighting definitions, subdimension
# mappings, and numerical thresholds for the GWRI.
#
# This file is the single source of truth for every research design constant.
# To change a weight, threshold, or classification rule, edit only this file.
# ============================================================================


# --- C1: Claim Specificity Scoring Criteria ----------------------------------
# Each environmental claim sentence is scored 0-4.
# The document-level C1 score is the average across all environmental sentences.


SPECIFICITY_CRITERIA = {
    0: (
        "No specific information - pure assertion only. "
        "Example: 'We care about the environment.'"
    ),
    1: (
```

```
        "Directional only - states a direction of change but no target or number. "
        "Example: 'We are working to reduce our emissions.'"
    ),
    2: (
        "Quantified - includes at least one numerical value. "
        "Example: 'We reduced emissions by 20%.'"
    ),
    3: (
        "Targeted - includes a number AND a named scope or comparison baseline. "
        "Example: 'We reduced Scope 1 CO2 by 20% compared to our 2019 baseline.'"
    ),
    4: (
        "Fully specified - includes a number, named scope, comparison baseline, "
        "AND a time reference. "
        "Example: 'We reduced Scope 1 CO2 emissions by 23% at our UK operations "
        "between 2020 and 2022, against a 2018 baseline.'"
    ),
}

# --- C4: Temporal Horizon Classification Rules ------------------------------
# Applied to future-tense environmental commitment sentences.

TEMPORAL_HORIZON_RULES = {
    "short_term": (
        "Deadline within approximately 1-2 years of the letter publication date. "
        "Typically a specific near-term year (e.g., next year, by 2024)."
    ),
    "medium_term": (
        "Deadline 3-5 years from the letter publication date. "
        "Example: 'by 2026', 'by 2028'."
    ),
    "long_term": (
        "Deadline 10 or more years from the letter date. "
        "Includes standard far-future target years: 2030, 2035, 2040, 2045, 2050, 2060."
    ),
    "vague": (
        "No deadline stated at all. Uses expressions such as: "
        "'in due course', 'over the coming years', 'eventually', "
        "'as soon as possible', 'in the near future'."
    ),
}

# Specific years that automatically classify a commitment as long-term
LONG_TERM_TARGET_YEARS = [2030, 2035, 2040, 2045, 2050, 2055, 2060]

# --- Subdimension Structure -------------------------------------------------
# Maps each of the 7 core variables to its conceptual subdimension.
# This is used by Module 6 (composite index calculation).

SUBDIMENSION_MAP = {
    "Substantive Disclosure Quality": ["C1", "C2"],
    "Commitment Accountability":      ["C3", "C4", "C6"],
    "Rhetorical Positioning":         ["C5", "C7"],
}

# --- Weighting - Version A (Theory-Based) -----------------------------------
# Subdimension weights reflect the relative theoretical importance of each
# dimension in the greenwashing literature.
# Variable weights reflect relative importance within each subdimension.
# All subdimension weights sum to 1.0.
```

```
# Variable weights within each subdimension sum to 1.0.
#
# Theoretical justification (to be expanded in thesis methodology):
#   - Substantive Disclosure Quality (0.40): specificity and evidence are the
#     most consistently validated signals in the greenwashing literature.
#   - Commitment Accountability (0.40): temporal and structural accountability
#     are central to the symbolic-vs-substantive commitment distinction.
#   - Rhetorical Positioning (0.20): important but less established empirically;
#     C7 (structural emphasis) is a newer contribution.


SUBDIMENSION_WEIGHTS_VERSION_A = {
    "Substantive Disclosure Quality": 0.40,
    "Commitment Accountability":      0.40,
    "Rhetorical Positioning":         0.20,
}


VARIABLE_WEIGHTS_VERSION_A = {
    # Within Substantive Disclosure Quality (must sum to 1.0)
    "C1": 0.60,   # Claim Specificity - most direct measure of claim quality
    "C2": 0.40,   # Lexical Inflation Risk - interaction-based, slightly less direct


    # Within Commitment Accountability (must sum to 1.0)
    "C3": 0.40,   # Action Temporality - strongest link to symbolic commitment
    "C4": 0.40,   # Commitment Horizon - equally central to accountability
    "C6": 0.20,   # Commitment Language Accountability - nuanced, harder to calibrate


    # Within Rhetorical Positioning (must sum to 1.0)
    "C5": 0.60,   # Promotional-to-Substantive Ratio - stronger literature support
    "C7": 0.40,   # Structural Emphasis Score - newer, less validated externally
}


# --- Weighting - Version B (Equal Weighting) ---------------------------------
# All 7 variables weighted equally (1/7 each).
# All 3 subdimensions weighted equally (1/3 each).
# Used as a robustness benchmark alongside Version A.


SUBDIMENSION_WEIGHTS_VERSION_B = {
    "Substantive Disclosure Quality": round(1 / 3, 10),
    "Commitment Accountability":      round(1 / 3, 10),
    "Rhetorical Positioning":         round(1 / 3, 10),
}


VARIABLE_WEIGHTS_VERSION_B = {
    "C1": round(1 / 7, 10),
    "C2": round(1 / 7, 10),
    "C3": round(1 / 7, 10),
    "C4": round(1 / 7, 10),
    "C5": round(1 / 7, 10),
    "C6": round(1 / 7, 10),
    "C7": round(1 / 7, 10),
}


# --- Winsorization -----------------------------------------------------------
# Before z-score standardization, each variable is winsorized so that extreme
# values are capped at these percentile bounds rather than excluded entirely.
# This limits the influence of extreme observations on the sample moments.


WINSORIZATION_PERCENTILES = (1, 99)
```

```
# --- Directional Alignment ------------------------------------------------
# Variables listed here have a higher raw score meaning LOWER greenwashing risk.
# They are multiplied by -1 after z-scoring so all variables are directionally
# consistent: higher standardized score = higher risk.

VARIABLES_TO_INVERT = ["C1"]
```

C.3 Rule-Based Variable Functions (modules/variables.py, abridged)

Implements C1 (Claim Specificity Index), C3 (Action Temporality Ratio), C4 (Commitment Horizon Distribution), and C7 (Structural Emphasis Score). Each function takes the list of environmental sentences from Module 3 and returns a document-level score. Long docstrings are omitted for brevity.

```
# modules/variables.py - Core variable scoring functions
# C1: Claim Specificity Index | C3: Action Temporality Ratio
# C4: Commitment Horizon Distribution | C7: Structural Emphasis Score

import re

# ── C1 patterns ──────────────────────────────────────────────────────────
_C1_QUANTITY   = re.compile(
    r'\b\d+(?:[.,]\d+)?\s*'
    r'(?:%|percent|tonne|ton|kg|mt|mwh|kwh|gwh|twh|mw|kw|gw|tw'
    r'|litre|liter|gallon|barrel|hectare|acre|km|km2|mile'
    r'|tree|plant|thousand|million|billion|m\b)', re.IGNORECASE)

_C1_STANDARD   = re.compile(
    r'\b(?:ISO\s*14001|ISO\s*50001|GRI(?:\s+Standards?)?|CDP(?:\s+[A-D](?:-list)?)?'
    r'|TCFD|SASB|UN\s*SDGs?|SDG\s*\d+|Paris\s+Agreement'
    r'|Science[- ]Based\s+Targets?|SBTi?|RE100|EV100|EP100)\b', re.IGNORECASE)

_C1_SCOPE      = re.compile(r'\bScope\s+(?:[123]|One|Two|Three)\b', re.IGNORECASE)
_C1_BASELINE   = re.compile(
    r'\b(?:vs\.?\s+\d{4}|compared\s+to|relative\s+to'
    r'|against\s+(?:a\s+)?(?:baseline|base\s+year)'
    r'|from\s+(?:a\s+)?\d{4}\s+baseline|baseline|base\s+year)\b', re.IGNORECASE)
_C1_TIME       = re.compile(
    r'\b(?:by\s+20\d{2}|in\s+20\d{2}'
    r'|between\s+20\d{2}\s+and\s+20\d{2}'
    r'|by\s+(?:end\s+of\s+)?(?:the\s+)?year\s+20\d{2})\b', re.IGNORECASE)
_C1_DIRECTIONAL = re.compile(
    r'\b(?:reduc(?:ing|e|ed)|decreas(?:ing|e|ed)|lower(?:ing|ed)?'
    r'|cut(?:ting)?|improv(?:ing|e|ed)|increas(?:ing|e|ed)'
    r'|invest(?:ing|ed)?|transition(?:ing|ed)?|deploy(?:ing|ed)?'
    r'|achiev(?:ing|e|ed)|working\s+towards?|committed\s+to)\b', re.IGNORECASE)

def _score_sentence_specificity(text: str) -> int:
    """Score 0-4 rubric: 4=quantity+scope/baseline+time, 3=quantity+scope/baseline,
    2=quantity only, 1=directional, 0=pure assertion."""
    has_evidence  = bool(_C1_QUANTITY.search(text)) or bool(_C1_STANDARD.search(text))
    has_scope     = bool(_C1_SCOPE.search(text))
    has_baseline  = bool(_C1_BASELINE.search(text))
    has_time      = bool(_C1_TIME.search(text))
    has_direction = bool(_C1_DIRECTIONAL.search(text))
```

```
    if has_evidence and (has_scope or has_baseline) and has_time: return 4
    elif has_evidence and (has_scope or has_baseline):            return 3
    elif has_evidence:                                            return 2
    elif has_direction:                                           return 1
    else:                                                         return 0

def compute_c1(environmental_sentences: list) -> dict:
    """Mean specificity score (0.0-4.0) across all environmental sentences."""
    if not environmental_sentences:
        return {"c1_score": None, "n_sentences_scored": 0, "sentence_scores": []}
    scores = [_score_sentence_specificity(s["text"]) for s in environmental_sentences]
    return {
        "c1_score":           round(sum(scores) / len(scores), 4),
        "n_sentences_scored": len(scores),
        "sentence_scores":    [{"sentence_id": s.get("sentence_id"),
                                "text": s["text"], "c1_score": sc}
                               for s, sc in zip(environmental_sentences, scores)],
    }

# ── C3 patterns ──────────────────────────────────────────────────────
_C3_FUTURE = re.compile(
    r'\b(?:will\s+\w+|shall\s+\w+|plan(?:s|ning)?\s+to'
    r'|commit(?:s|ted|ting)?\s+to|aim(?:s|ing)?\s+to'
    r'|intend(?:s|ed|ing)?\s+to|going\s+to|target(?:s|ing|ed)?\s+to'
    r'|seek(?:s|ing)?\s+to|set(?:ting)?\s+out\s+to)\b', re.IGNORECASE)

_C3_COMPLETED = re.compile(
    r'\b(?:reduced|decreased|lowered|cut|improved|increased|grew'
    r'|invested|transitioned|deployed|installed|achieved|met|completed'
    r'|delivered|reached|launched|implemented|certified|verified|eliminated)\b',
    re.IGNORECASE)

_C3_NEGATION = re.compile(
    r'\b(?:will\s+not\s+\w+|shall\s+not\s+\w+|will\s+never\s+\w+'
    r'|cannot\s+commit\s+to)\b', re.IGNORECASE)

def compute_c3(environmental_sentences: list) -> dict:
    """Action Temporality Ratio = n_future / (n_completed + n_ongoing).
    Returns None with all_future=True when denominator is zero."""
    if not environmental_sentences:
        return {"c3_score": None, "all_future": False,
                "n_future": 0, "n_completed": 0, "n_ongoing": 0,
                "classified_sentences": []}
    n_future = n_completed = n_ongoing = 0
    classified = []
    for s in environmental_sentences:
        if _C3_FUTURE.search(s["text"]) and not _C3_NEGATION.search(s["text"]):
            temporality = "future";   n_future += 1
        elif _C3_COMPLETED.search(s["text"]):
            temporality = "completed"; n_completed += 1
        else:
            temporality = "ongoing";   n_ongoing += 1
        classified.append({"sentence_id": s.get("sentence_id"),
                           "text": s["text"], "position": s.get("position"),
                           "temporality": temporality})
    denom = n_completed + n_ongoing
    if denom == 0 and n_future > 0:
        return {"c3_score": None, "all_future": True,
                "n_future": n_future, "n_completed": 0, "n_ongoing": 0,
```

```
                "classified_sentences": classified}
    return {"c3_score": round(n_future / denom, 4) if denom > 0 else None,
            "all_future": False, "n_future": n_future,
            "n_completed": n_completed, "n_ongoing": n_ongoing,
            "classified_sentences": classified}
# ── C4 pattern ──────────────────────────────────────────────────────────
_C4_YEAR = re.compile(r'\b(20[0-9]{2})\b')


def _classify_horizon(text: str, letter_year: int) -> str:
    """Classify commitment horizon relative to letter year.
    offset ≤2 → short_term | 3-9 → medium_term | ≥10 → long_term | no year → vague"""
    matches = _C4_YEAR.findall(text)
    if not matches or letter_year == 0:
        return "vague"
    offset = max(int(y) for y in matches) - letter_year
    if offset < 0:   return "vague"
    if offset <= 2:  return "short_term"
    if offset <= 9:  return "medium_term"
    return "long_term"


def compute_c4(classified_sentences: list, letter_year: int = 0) -> dict:
    """C4 = (n_long_term + n_vague) / n_total_future. Scores future sentences only."""
    future = [s for s in classified_sentences if s.get("temporality") == "future"]
    if not future:
        return {"c4_score": None, "n_short_term": 0, "n_medium_term": 0,
                "n_long_term": 0, "n_vague": 0, "n_total_future": 0,
                "horizon_sentences": []}
    counts = {"short_term": 0, "medium_term": 0, "long_term": 0, "vague": 0}
    horizon_sentences = []
    for s in future:
        h = _classify_horizon(s["text"], letter_year)
        counts[h] += 1
        horizon_sentences.append({"sentence_id": s.get("sentence_id"),
                                  "text": s["text"], "horizon": h})
    n = len(future)
    return {"c4_score":          round((counts["long_term"] + counts["vague"]) / n, 4),
            "n_short_term":      counts["short_term"],
            "n_medium_term":     counts["medium_term"],
            "n_long_term":       counts["long_term"],
            "n_vague":           counts["vague"],
            "n_total_future":    n,
            "horizon_sentences": horizon_sentences}


# ── C7 ──────────────────────────────────────────────────────────────────


def compute_c7(environmental_sentences: list) -> dict:
    """C7 = mean document position of environmental sentences (0.0-1.0).
    0.0 = foregrounded; 1.0 = relegated to end."""
    if not environmental_sentences:
        return {"c7_score": None, "n_environmental_sentences": 0, "position_values": []}
    positions = [s["position"] for s in environmental_sentences]
    return {"c7_score":                  round(sum(positions) / len(positions), 4),
            "n_environmental_sentences": len(positions),
            "position_values":           positions}
```

C.4 Composite Score Assembly (modules/composite.py, abridged)

Assembles the GWRI composite from the four rule-based variables via a five-step pipeline: risk adjustment, winsorization, all-future imputation for C3, z-score standardization, and

weighted aggregation under Version A and Version B weights. Private helper functions (_winsorize_variable, _zscore_variable, etc.) are omitted; the full source is available from the authors.

```
# modules/composite.py - GWRI Composite Score (key logic only)
# Full pipeline: risk-adjust → winsorise → impute → z-score → composite
# Helper functions (_percentile, _winsorise_variable, _zscore_variable,
# _weighted_score, _is_valid_for_gwri) omitted for brevity.


import math
from config.rules import WINSORIZATION_PERCENTILES


_VARS            = ["c1", "c3", "c4", "c7"]
_WINSOR_LOW_PCT  = WINSORIZATION_PERCENTILES[0]   # 1st percentile
_WINSOR_HIGH_PCT = WINSORIZATION_PERCENTILES[1]   # 99th percentile


# Version A - theory-based weights (40/40/20 subdimension structure) WEIGHTS_VERSION_A =
{"c1": 0.40, "c3": 0.20, "c4": 0.20, "c7": 0.20}


# Version B - equal weights across the four partial-GWRI variables WEIGHTS_VERSION_B =
{"c1": 0.25, "c3": 0.25, "c4": 0.25, "c7": 0.25}


def build_composite(observations: list) -> list:
    """
    Build GWRI composite scores from Module 4 outputs.


    Pipeline per observation:
      1. Risk-adjust: invert C1 (high specificity = low risk → subtract from 4)
      2. Winsorise at 1st/99th percentile of finite sample values
      3. Impute all_future documents: assign upper C3 winsorisation bound
      4. Z-score standardise (sample std, Bessel correction)
      5. Validity check: requires c1, c7, and at least one of c3/c4
      6. Weighted composite - Version A (equal) and Version B (theory-based)
         Weights renormalise over available variables when any are missing.


    Returns enriched list with gwri_version_a, gwri_version_b added to each doc.
    """
    if not observations:
        return []


    # Step 1 - Risk-adjust
    # C1 inverted: higher specificity = lower risk; all others: higher = higher risk
    risk = {v: [] for v in _VARS}
    for obs in observations:
        c1 = obs.get("c1_score")
        risk["c1"].append(4.0 - c1 if c1 is not None else None)
        risk["c3"].append(obs.get("c3_score"))
        risk["c4"].append(obs.get("c4_score"))
        risk["c7"].append(obs.get("c7_score"))


    # Steps 2-4 - Winsorise → impute all_future C3 → z-score
    # (see helper functions in full source)
    winsor, winsor_bounds = {}, {}
    for v in _VARS:
        w_vals, lo, hi  = _winsorize_variable(risk[v])
        winsor[v]       = w_vals
        winsor_bounds[v] = (lo, hi)
```

```
    c3_upper = winsor_bounds["c3"][1]
    for i, obs in enumerate(observations):
        if obs.get("all_future") is True and winsor["c3"][i] is None:
            if c3_upper is not None:
                winsor["c3"][i] = c3_upper


    z, zv_flags = {}, {}
    for v in _VARS:
        z[v], zv_flags[v] = _zscore_variable(winsor[v])


    # Steps 5-6 - Validity check + weighted composite
    results = []
    for i, obs in enumerate(observations):
        doc = dict(obs)
        z_dict = {v: z[v][i] for v in _VARS}
        winsor_avail = {v: winsor[v][i] for v in _VARS}
        doc["valid_for_gwri"] = _is_valid_for_gwri(winsor_avail)


        if doc["valid_for_gwri"]:
            doc["gwri_version_a"], doc["weights_version_a"] = \
                _weighted_score(z_dict, WEIGHTS_VERSION_A)
            doc["gwri_version_b"], doc["weights_version_b"] = \
                _weighted_score(z_dict, WEIGHTS_VERSION_B)
        else:
            doc["gwri_version_a"] = doc["gwri_version_b"] = None


        results.append(doc)
    return results
```

Note: C2 (Lexical Inflation Risk), C5 (Promotional vs Substantive Framing), and C6 (Commitment Language Accountability) are computed via AI-assisted scoring using the prompt templates in config/prompts.py and the functions in modules/ai_variables.py. These are included in the full source available upon request.

Appendix D - GWRI: Variable Definitions and Construction Formula

The Greenwashing Warning Risk Index (GWRI) is a composite index constructed from seven sub-indicators (C1-C7) that measure different dimensions of greenwashing rhetoric in CEO letters. This appendix defines each variable and sets out the full construction formula. The current thesis implements four rule-based variables (C1, C3, C4, C7), labelled partial GWRI. The remaining three AI-assisted variables (C2, C5, C6) are included in the theoretical design and the full code but are deferred to future research.

D.1 GWRI Sub-Indicator Summary (C1-C7)

Wt. A (full) = effective weight in the full seven-variable Version A (theory-based subdimension × within-subdimension weight). **Wt. A (partial)** = weight used in the partial GWRI implemented in this thesis (Version A, renormalised over C1, C3, C4, C7 only). - = variable coded but excluded from the partial GWRI used in this thesis; included in the full theoretical design.

Table D1: GWRI Sub-Indicator Summary

Var.	Full Name	Subdimension	What It Measures / Scoring Logic	Range	Method	Wt. A (full)	Wt. A (partial)

C1	Claim Specificity Index	Substantive Disclosure Quality	Mean specificity of environmental sentences. Scored 0-4: 0 = pure assertion; 1 = directional; 2 = quantified; 3 = quantified + scope/baseline; 4 = fully specified with time reference. Inverted before aggregation.	0-4	Rule-based	0.24	0.400
C2	Lexical Inflation Risk	Substantive Disclosure Quality	Mean inflation score of environmental sentences matching the buzzword lexicon. 0 = substantive despite buzzword; 0.5 = mild inflation; 1 = strong inflation. Only triggered sentences enter the document mean.	0-1	AI-assisted	0.16	-
C3	Action Temporality Ratio	Commitment Accountability	Ratio of future-oriented to completed and ongoing environmental sentences: n_future / (n_completed + n_ongoing). Higher ratio = more aspirational, less action-oriented language.	0-∞	Rule-based	0.16	0.200
C4	Commitment Horizon Distribution	Commitment Accountability	Proportion of future commitments classified as long-term (≥10 years from letter date) or vague (no deadline stated): (n_long + n_vague) / n_future. Higher = harder to hold accountable.	0-1	Rule-based	0.16	0.200
C6	Commitment Language Accountability	Commitment Accountability	Mean accountability score of forward-looking environmental sentences. 0 = accountable (clear, specific, bounded); 0.5 = partially accountable; 1 = weak/evasive (heavily hedged or no verifiable obligation).	0-1	AI-assisted	0.08	-
C5	Promotional vs Substantive Framing	Rhetorical Positioning	Mean framing score across all environmental sentences. 0 =	0-1	AI-assisted	0.12	-

			substantive (factual, measurable); 0.5 = mixed; 1 = promotional (emotional, aspirational, brand-identity language with no verifiable claim).				
C7	Structural Emphasis Score	Rhetorical Positioning	Mean normalised document position of environmental sentences, where 0.0 = first sentence and 1.0 = last sentence. Low score = environmental content foregrounded; high score = relegated to end of document.	0-1	Rule-based	0.08	0.200

Subdimension weights (Version A): Substantive Disclosure Quality = 0.40; Commitment Accountability = 0.40; Rhetorical Positioning = 0.20. Version B assigns equal weight 1/7 ≈ 0.143 to each variable. C6 is listed under Commitment Accountability but between C4 and C5 for display clarity.

D.2 GWRI Construction Pipeline

The index is computed in four sequential steps applied to each firm-year document:

Step 1 - Variable scoring. Each sub-indicator C_i is computed from the set of environmental sentences detected in the CEO letter. Rule-based variables (C1, C3, C4, C7) use regular-expression matching. AI-assisted variables (C2, C5, C6) use a structured prompt sent to a language model scorer.

Step 2 - Risk adjustment. C1 is inverted so that higher values uniformly indicate higher greenwashing risk:

$$C1_{risk} = 4 - C1_{raw}$$

All other variables are already risk-aligned (higher raw score = higher greenwashing risk).

Step 3 - Winsorisation and z-score standardisation. Each risk-adjusted variable is first winsorised at the 1st and 99th sample percentiles, then standardised across all N firm-year observations:

$$\tilde{z}(C_i) = C_{i,winsor} - \mu_i \; / \; \sigma_i$$

where μ_i = sample mean, σ_i = sample standard deviation (Bessel's correction, n−1)

Documents where all environmental sentences are future-oriented (*all_future* = True) receive an imputed C3 value equal to the upper winsorisation bound before standardisation, reflecting maximum temporal risk.

Step 4 - Weighted composite. Standardised variables are aggregated into the composite index using two weighting schemes:

Version A - Partial GWRI (theory-based, implemented in thesis):

$$\text{GWRI-A} = 0.40 \cdot \tilde{z}(C1) + 0.20 \cdot \tilde{z}(C3) + 0.20 \cdot \tilde{z}(C4) + 0.20 \cdot \tilde{z}(C7)$$

Version B - Partial GWRI (equal-weighted, robustness check):

$$\text{GWRI-B} = 0.25 \cdot \tilde{z}(C1) + 0.25 \cdot \tilde{z}(C3) + 0.25 \cdot \tilde{z}(C4) + 0.25 \cdot \tilde{z}(C7)$$

Full GWRI-A (theoretical design, all seven variables):

$$\begin{aligned}\text{GWRI-A}_{\text{full}} = {} & 0.40 \cdot [0.60 \cdot \tilde{z}(C1) + 0.40 \cdot \tilde{z}(C2)] \\ & + 0.40 \cdot [0.40 \cdot \tilde{z}(C3) + 0.40 \cdot \tilde{z}(C4) + 0.20 \cdot \tilde{z}(C6)] \\ & + 0.20 \cdot [0.60 \cdot \tilde{z}(C5) + 0.40 \cdot \tilde{z}(C7)]\end{aligned}$$

Positive GWRI scores indicate above-average greenwashing rhetoric risk relative to the sample. Negative scores indicate below-average risk. The index is sample-standardised and has no natural absolute scale.

When a variable is missing for a given document, weights are renormalised proportionally over the available variables so that they continue to sum to 1. A document is eligible for a GWRI score only when C1, C7, and at least one of C3 or C4 are non-missing (minimum three of four partial variables present).



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